Characteristic-Data Bounds for Multiple Context-Free Languages under Fixed Finite-Semigroup Typing

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Abstract

We investigate the total encoded length $\|K\|_+ = \sum_{w\in K}(|w|+1)$ of characteristic positive data for multiple context-free languages (MCFLs) that are tuple-substitutable under a fixed finite-semigroup typing. Qualitative reconstruction and positive-data identification are inherited from Yoshinaka's distributional framework; our contribution is quantitative. The alphabet, fan-out, rule-rank cap and typing are fixed throughout the corresponding bounds. For a reduced good MCFG G of fixed fan-out and rule rank, with rule thickness $\bar{\theta}_G$, we obtain $O(\|G\|^2\bar{\theta}_G)$ total characteristic-data length when the finite typing is determined by a prefix–suffix window, optionally together with values in an arbitrary finite group. This includes noncommutative groups: a new owner-wise block-replacement lemma shortens several tuple components simultaneously without commuting their ordered products.

The sufficient typing condition extends to all local-group positive-image semigroups, with bound $O(\|G\|^3\bar{\theta}_G)$, via a sliding-factor group representation. For arbitrary standard input MCFGs (including deleting rules and empty components), the bounds become $O(n^3t)$ and $O(n^4t)$, where n is the input grammar size and t records the shortest derivation for each realizable nonempty-component pattern. A separate conservative learner achieves Gold identification; the length bounds do not imply a polynomial convergence stage. We give an intrinsic fan-out-two, rank-two non-context-free example, and a cubic bound for rank-one input grammars under arbitrary fixed typing. Higher-fan-out superpolynomial characteristic-data lower bounds remain open.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-semigroup typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be recovered from an enumeration of its positive examples alone [1, 2]. Distributional approaches make this possible for restricted grammar classes by turning observed interchangeability of substrings or tuples into grammar nonterminals. For context-free grammars (CFGs) this includes substitutable languages [3] and Yoshinaka's (k, ℓ) hierarchy [12]. Yoshinaka extended the idea to multiple context-free grammars (MCFGs), whose nonterminals generate tuples and whose substitutions are tested by multi-hole sentence contexts [13,14]. MCFGs are a standard mildly context-sensitive formalism for discontinuous and cross-serial dependencies [29, 30].

Two words, finite types, and long characteristic data. The elementary language $L_1 = \{\mathtt{a}, \mathtt{aa}\}$ already explains why a finite comparison bias matters. Its two words share the empty context,

yet are not freely substitutable; their lengths have different parity. More generally, on $\Sigma = \{\mathtt{a}\}$ let $L_m = \{\mathtt{a}, \mathtt{a}^{2^m}\}$ for $m \geq 1$ and fix the *same* parity typing $s(\mathtt{a}^k) = k \bmod 2$ for all m. Although untyped substitutability fails, every L_m is (f, s)-tuple-substitutable for every f: equal-type tuples sharing an accepting context must produce the same one of the two words, and over a unary alphabet their replacement consequently agrees in every context. This is the finite bias at work: it prohibits unjustified identifications without itself encoding how large the exponent 2^m is (cf. [25]).

The same two-word family exposes the quantitative question. For our sound, sample-consistent reconstruction operator, every characteristic sample for L_m must contain *both* words: either singleton datum alone also belongs to a singleton target, on which soundness prevents the reconstruction of the missing word. Thus the characteristic set has cardinality only two, but its total encoded length is $2^m + 3$ under $\|K\|_+ = \sum_{w \in K}(|w| + 1)$. Nevertheless, L_m has a CFG with $m + O(1)$ rules, using $A_0 \to \mathtt{a}$, $A_{i+1} \to A_i A_i$ and $S \to \mathtt{a} \mid A_m$. Its rule thickness also grows with 2^m, so this example motivates rather than contradicts our thickness-sensitive bounds: *few characteristic words need not mean short characteristic data.*

The issue is not confined to context-free languages. For example, the non-context-free fan-out-two language $T = \{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$ belongs to $\mathcal{C}^{\mathrm{mcf}}_{2,s_{1,1}}$, where $s_{1,1}$ records the first and last letter. Indeed, $\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$ is strictly 2-local, while the two equal-count conditions are an abelian kernel; the later Lemma 5.2 and Proposition 2.9 give typed substitutability. Thus finite typing restricts permitted substitutions, whereas MCFGs express the unbounded dependencies. Yoshinaka's signature theory [15] provides the qualitative learning principle; our task is to bound the *total length* of the positive witnesses.

This paper fixes, as part of the learning bias, a surjective finite-semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ and compares observed *nonempty* tuple components only when their s-values agree. The canonical fresh-identity completion $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ is used only to evaluate grammar templates, including empty terminal gaps. The typing and the fan-out bound are fixed before any data are seen, separating the choice of a finite bias from learning under that bias [5]; prefix–suffix windows and DFA transition morphisms are typical instances. Other biases restrict derivations by regular control sets [6, 7] or restrict the evidence used to compare factors [9, 11]; positive-data learning of other mildly context-sensitive formalisms is studied in [19], and MCFG learning with membership queries or dual context sets in [16, 18, 20]. A recent substitution lemma for MCFLs [23] is a necessary condition for membership, not a learning criterion.

Prior work and scope. The qualitative observed-tuple grammar and identification mechanism derive from Yoshinaka's MCFG work [13,14]; his finite-signature framework also covers bounded-rank learning [15]. Its main proof concerns the context-indexed $\mathrm{SUBST}_{\mathrm{D}}$ constructor. Our concrete Const/Link/Comp grammar is verified directly below; we do not assume a rule-by-rule identification with Yoshinaka's $\mathrm{SUBST}_{\mathrm{P}}$ implementation. Polynomial characteristic data for prefix–suffix typed CFGs are due to Yoshinaka [12]. The companion CFG paper [25] treats fixed finite-semigroup typing and proves a fan-out-one lower bound for a fixed two-element semilattice typing [25, Proposition 6.4]. Its arXiv version 5 is identical to the revised manuscript submitted to Theoretical Computer Science. All upper bounds in the present paper are proved independently of that companion lower bound; only the specified negative boundary consequence invokes it. In Yoshinaka's (k, ℓ)-substitutable CFG construction [12, Lemmas 2, 6 and 7], Lemma 6 bounds the *number* of witnesses by $|P'|^{7(k+\ell)+1}$ for a converted Greibach-normal-form grammar G' with rules P'; Lemma 7 separately bounds the length of each word by a polynomial in the converted grammar size and its thickness. In particular, the cardinality bound alone is not a total-encoded-length bound. Our displayed $O(\|G\|^2\bar{\theta}_G)$ bound instead measures total encoded length relative to a *reduced good* presentation G, and its degree

two in $\|G\|$ is independent of fixed k, ℓ. The hidden constant contains the finite type space, which may have size $|\Sigma|^{O((k+\ell)fr_{\max})}$ when Σ varies. Different normal forms and size parameters prevent a direct claim of improved polynomial degree over Yoshinaka's result at fan-out one; none is intended. The comparison concerns stated, presentation-specific upper bounds and not a bound on the optimal sample size. The rank-one separation uses the non-branching/finite-index EDT0L correspondence in Gallot [21], with the published word-problem obstruction of Bishop et al. [22]. The qualitative MCFG learning mechanism, polynomial hypothesis updates, and finite characteristic-set *cardinality* are not independent novelties of this paper. The first independent quantitative input is the simultaneous shortening lemma for ordered products of blocks with common owners (Proposition 4.10); the resulting window–group witness bound is Theorem 4.14. The second is the sliding-factor representation of local-group typings (Theorem 4.28), which yields the more general Theorem 4.24. The third is the transfer to arbitrary standard MCFG presentations using component-pattern thickness (Proposition 4.34). The intrinsic fan-out/rank example and the high-arity obstruction explain the scope, without supplying matching lower bounds.

Contributions. Our object is the *total encoded length* of characteristic positive data relative to an MCFG presentation, for typed *tuples* at higher rule rank, where several component boundaries must be preserved simultaneously. An untyped pruning argument cannot simply shorten child derivations while retaining their types: in noncommutative finite groups, blocks from one subtree may appear in different orders across different output components. The new quantitative step is an *owner-wise simultaneous* selection of replacements preserving all ordered root products, rather than preserving each omitted subtree's group value separately. The bounds below hold with $f, r_{\max}, |\Sigma|$ and the typing fixed; their hidden constants can grow rapidly with these fixed parameters. For a reduced good MCFG G put $\bar{\theta}_G = \max\{1, \theta_G\}$, where θ_G is rule thickness (Section 4). With alphabet, fan-out, rule-rank bound and typing fixed:

Fixed typing regime	Positive-data length	Source
Rule rank one; arbitrary typing	$O(\|G_{\text{in}}\|^3)$	Thm. 4.6
Prefix–suffix window	$O(\|G\|^2\bar{\theta}_G)$	Thm. 4.8
Window and any finite group	$O(\|G\|^2\bar{\theta}_G)$	Thm. 4.14
Positive image in **LG**	$O(\|G\|^3\bar{\theta}_G)$	Thm. 4.24

The first table row concerns a rank-one input representation; the other three rows concern reduced good presentations. Here $\|G\|$ is a presentation parameter, not an intrinsic minimal grammar size, and $\bar{\theta}_G$ measures the shortest successful derivation visiting a rule. In particular none of the displayed bounds is polynomial in grammar size alone without controlling thickness. The group result includes noncommutative groups; the abelian case has sharper constants (Theorem 4.15). Corollary 4.35 removes all good-form restrictions on the *input* MCFG: its length bounds are $O(n^3t)$ and $O(n^4t)$, respectively, in the input size n and component-pattern thickness t. The **LG** hypothesis concerns the positive-image semigroup; its known algebraic characterization [36] is distinct from the learning bound. Even at fan-out one, our local-group bound strictly extends the companion CFG paper's sufficient condition $\mathbf{LI} \vee \mathbf{Grp}$ (Proposition 4.29). This is a consequence of the new MCFG shortening argument, not a claim that the companion already established the local-group theorem. A companion-dependent pseudovariety-level lower-bound consequence is recorded only at fan-out one (Corollary 4.31); its negative direction is not self-contained in the present paper. We also give a computable rule-only data budget (Proposition 4.37), a language-based bound for abelian-filtered targets (Theorem 4.39), and a boundary-typed non-context-free target of intrinsic fan-out and rule rank two (Theorem 5.3). Characteristic data concern language-level exactness of a set-driven operator; syntactic stabilization is supplied by a separate conservative learner.

Section 2 fixes the model, Section 3 the reconstruction and its exactness, Section 4 the quantitative bounds, Section 5 the separations, and Section 6 the higher-arity obstruction. Supplementary forcing-depth and finite-target arguments are collected in Appendix E to separate ancillary restrictions from the quantitative upper-bound proofs.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word, and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly.

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; thus $\perp, \perp, \ldots$ is the text for $\emptyset$. The *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$, and it TxtEx-identifies a class if it does so for each member. A learner is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$. For a language L, a finite $\mathcal{F} \subseteq L$ is a *characteristic sample for* $\mathcal{B}$ if $\mathcal{F} \subseteq K \subseteq L$ implies $L(\mathcal{B}(K)) = L$ for every finite K (cf. [4]). Unlike TxtEx identification this is a statement about generated languages: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

For finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$.

2.2 Fixed Finite-Semigroup Typing

Following the companion CFG paper [25], the learning bias is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by its multiplication table and the letter images (so $s(w)$ is computable in linear time). A finite codomain can always be replaced by the positive image, so surjectivity loses no generality.

Definition 2.3 (Fresh completion, tuple types, refinement)**.** Let $\mathsf{S}^\sharp = \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity (even if S already has one). The canonical monoid extension $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ satisfies $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}(w) = s(w)$ for $w \in \Sigma^+$. For $\vec{w} = (w_1, \ldots, w_d)$ write $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \ldots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ it agrees with $s^{(d)}$. The fresh identity evaluates empty template factors but does not create an additional positive type. For typings $s, t : \Sigma^+ \to$ finite semigroups write $s \preceq t$ if $\ker t \subseteq \ker s$ on Σ^+, and (s, t) for their product restricted to its positive image.

Positive typing and auxiliary morphisms. Empty terminal gaps in rule templates are evaluated by $\widehat{s}$ with values in $\mathsf{S}^\sharp$. An auxiliary finite-monoid morphism $q : \Sigma^* \to N$ represents the positive typing $q|_{\Sigma^+}$, and Lemma C.1 proves equivalence with its fresh-identity completion on all nonempty components and all successful trimmed refinements. In particular, algebraic conditions such as **LG** concern the positive image S, not $\mathsf{S}^\sharp$. Write s_{triv} for the one-element positive typing.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k+\ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v \rangle, \langle u', v' \rangle$ define

$$x \cdot y := \widehat{s}_{k,\ell}(xy), \qquad x \cdot \langle u, v \rangle := \langle \mathrm{pref}_k(xu), v \rangle,$$
$$\langle u, v \rangle \cdot y := \langle u, \mathrm{suf}_\ell(vy) \rangle, \quad \langle u, v \rangle \cdot \langle u', v' \rangle := \langle u, v' \rangle.$$

A case check gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$; since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, associativity is inherited from concatenation and $\widehat{s}_{k,\ell}(\varepsilon)$ is the identity. Thus $\widehat{s}_{k,\ell}$ is a finite-monoid morphism; its positive restriction $s_{k,\ell}$ is the boundary typing, and since the short-word tag separates ε from Σ^+, $\widehat{s}_{k,\ell}$ is isomorphic to $\widehat{s_{k,\ell}}$. A positive typing s with $s \preceq s_{k,\ell}$ is called (k,ℓ)-*boundary-determined*, or briefly a *boundary typing*.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation)**.** An MCFG [29, 30] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a constant rule. The tuple languages $L_A(G)$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$.

We write $\mathsf{MCFL}(f, r)$ for the languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$, so that f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$. Minimum rule rank of a language refers to all presentations, not to a given grammar. $\|G\|$ is the length of a fixed symbol-level encoding counting nonterminal and rule records and every terminal or variable occurrence in templates. For nonterminals of equal fan-out d, $A \to B$ abbreviates the identity-template rule $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font.

Fan-out and rule rank are distinct resources: there is no rank reduction preserving an arbitrary fixed fan-out [24, 32]. The reconstruction below therefore imposes neither a binary normal form nor a known rank cap.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity* d is a word $E = v_0 \Box_1 v_1 \cdots v_{d-1} \Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \ldots, v_{d-1} \in \Sigma^+$; such contexts are *admissible*, and $\mathcal{E}_d$ denotes their set ([14, Section 3.1]). $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^d |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$.

Definition 2.6 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions ${}_{⓪}a_{1①} \cdots a_{n⓷}$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \leq i_1 < i_1' < i_2 < i_2' < \cdots < i_d < i_d' \leq n$; it determines $\vec{x} = (w[i_1 : i_1'], \ldots, w[i_d : i_d']) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \ldots, \Box_d$, so that $E[\vec{x}] = w$. For a finite K, $\mathrm{Obs}_d(K)$ is

the set of tuples $\vec{x}$ arising from occurrences in words of K, i.e. $\vec{x} \in (\Sigma^+)^d$ with $E[\vec{x}] \in K$ for some $E \in \mathcal{E}_d$, and $\mathrm{Obs}(K) := \bigcup_{d \le f} \mathrm{Obs}_d(K)$.

For $L \subseteq \Sigma^*$ and $\vec{x} \in (\Sigma^*)^d$ the *d-tuple distribution* is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$.

Definition 2.7 ((f,s)-tuple substitutability)**.** Let $f \ge 1$ and $s : \Sigma^+ \to \mathsf{S}$ a finite-semigroup morphism. A language $L \subseteq \Sigma^*$ is (f,s)-*tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

Empty components are never compared (Yoshinaka's convention [14, Section 3.1 and footnote 3]; see [8] for CFGs); $\varepsilon \in L$ is handled at the start symbol. The restriction matters already at fan-out one: $\mathtt{a}$ and ε share a context in $\mathtt{a}^+$ but have different distributions. At fan-out one, $\mathcal{D}_L^{(1)}(x)$ is the two-sided distribution $\{(u,v) : uxv \in L\}$. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,s}^d \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $\widehat{s}^{(d)}(\vec{x}) = \widehat{s}^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Definition 2.8 (The target class)**.** $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ is the class of (f,s)-tuple-substitutable f-MCFLs. A target in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ for a boundary typing $\widehat{s}$ is called *boundary-typed*.

If $s \preceq t$, then $\mathcal{C}_{f,s}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,t}^{\mathrm{mcf}}$, since equal finer types are equal coarser types; in particular $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$. The class is not monotone in f: only $\mathcal{C}_{f+1,s}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$ holds, and already for trivial typing $\{w\#w^R\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5]. Thus f is part of the bias. For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$: it is regular, and equal componentwise types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and $\widehat{s}$.*

(i) If L_1, L_2 are (f,s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f,s)-tuple-substitutable, A is an abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a monoid morphism, $\mathsf{H} \le \mathsf{A}$ and $\gamma \in \mathsf{A}$, then $L \cap \varphi^{-1}(\gamma + \mathsf{H})$ is (f,s)-tuple-substitutable.

Whenever the intersection is an f-MCFL, it belongs to $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.

Proof. (i) A context accepting equal-type $\vec{x}, \vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If a context accepts $\vec{x}, \vec{y}$ in $\varphi^{-1}(\gamma + \mathsf{H})$, then $\delta := \sum_i (\varphi(y_i) - \varphi(x_i)) \in \mathsf{H}$; by commutativity $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}]) = \delta$ for every E', so coset membership transfers in both directions. Thus $\varphi^{-1}(\gamma + \mathsf{H})$ is (f, s_{triv})-, hence (f,s)-tuple-substitutable; apply (i). □

Consequently every f-MCFL of the form $\widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ lies in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$; such factorizations are not supplied to the learner.

Notation conventions. Throughout, K denotes a finite *sample* and G a grammar; S is the fixed positive-image semigroup. In the quantitative sections, θ_G is rule thickness (the shortest *sentence* length witnessing each rule), t_G is nonterminal thickness, and $\|G\|$ is encoding size. The auxiliary symbols A and N denote a finite group and a finite monoid respectively when stated locally; T is a derivation tree unless newly bound in a local algebraic proof. In the effective-witness subsection, $\lambda(X)$ and $\mu(X)$ denote respectively minimum inside yield length and minimum outside context length; $\lambda_{k,g}$ in the local-group section is a separately named *typing*, not either length. All such temporary notation is scoped to its displayed statement or proof.

3 Yoshinaka's Learning Principle and Quantitative Witnesses

Fixed-rank learning is a specialization of Yoshinaka [15]. We first identify the relevant signature and its concrete implementation, retaining below only those invariants and finite witnesses required by the *total-length* analysis in Section 4 and the separate unbounded-rank extension.

3.1 Set-Driven Reconstruction

Auxiliary good form for the proofs. A rule is *nondeleting* if every variable occurs exactly once; this alone does not forbid empty output components. A nondeleting rule is *nonpermuting* if, for each j, $\xi_j^1, \dots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$; following [14, Section 2.3] it is λ-*free* if no α_i is empty, and a nonpermuting rule is *nonmerging* if no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. A rule, template or presentation is *good* if it is λ-free, nondeleting, nonpermuting and nonmerging. By induction on derivations, every component of every tuple derived from a nonterminal of a λ-free grammar is nonempty.

All interaction between templates and contexts below rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \dots, \alpha_e)$ be a good template of rank r with child arities $d_1, \dots, d_r$, let $E \in \mathcal{E}_e$, let $1 \le j \le r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \neq j$. Let E_j be the word obtained from $E[(\alpha_1, \dots, \alpha_e)]$ by replacing ξ_j^κ by $\square_\kappa$ and every other variable ξ_i^ι by $u_{i,\iota}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \dots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \dots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Proof. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \dots, d_j$, since E places $\alpha_1, \dots, \alpha_e$ from left to right. Consider ξ_j^κ and $\xi_j^{\kappa+1}$. If both lie in one α_i, nonmerging puts a terminal or a variable ξ_i^ι with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Definition 3.2 (Composition witness)**.** Let K be finite and $r \ge 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \dots, \vec{u}_r))$ with $\vec{z} \in \mathrm{Obs}(K)$, ρ a good template of rank r whose output and child arities are at most f, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r)$.

Since each $\alpha_i(\vec{u}_1, \dots, \vec{u}_r) = z_i$ fixes the positions of all child components inside $\vec{z}$, a composition witness is the same as a segmentation of an observed occurrence of $\vec{z}$ into terminal gaps and labelled child intervals. If $E[\vec{z}] \in K$ with $E \in \mathcal{E}_e$, Lemma 3.1 shows that every $\vec{u}_j \in \mathrm{Obs}(K)$. Linearity and nondeletion make the children contribute disjointly to $\vec{z}$, and each contributes at least one letter, so

$$r \le \textstyle\sum_j \|\vec{u}_j\|_1 \le \|\vec{z}\|_1 \le \max_{w \in K} |w| \le \|K\|_+. \tag{3.1}$$

Definition 3.3 (Capped and uncapped reconstruction operators)**.** Fix f and a finite-semigroup typing s with canonical completion $\widehat{s}$. For $r_{\max} \ge 1$ let $\mathcal{B}_{f,s}^{\le r_{\max}}(K) = \widehat{G}_{\le r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $d \le f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every observed $\vec{x}$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$ for every composition witness of rank $1 \le r \le r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for observed $\vec{x}, \vec{y}$ of the same arity d with $\widehat{s}^{(d)}(\vec{x}) = \widehat{s}^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\leq \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, by $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. Every $[\vec{x}]$ derives only tuples with nonempty components, since Comp templates are λ-free; so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start are exactly the rules of Types I, II and III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ is the operator of Yoshinaka's unbounded-rank learner $\mathcal{A}(p, *)$ [14, Corollary 2].

3.2 Qualitative learning as an inherited principle

For fixed f, r, s, finite-sort specialization of Yoshinaka's signature theorem [15, Theorem 1] gives qualitative positive-data identification and polynomial hypothesis updates (Corollary B.3). The relation to Yoshinaka's signature formulation is described in Appendix B; soundness and completeness for the actual Const/Link/Comp operator are established by direct arguments, without appealing to equivalence with $\mathrm{SUBST}_{\mathrm{P}}$. No total-length bound is inherited from characteristic-set *cardinality*: this is the new quantitative question.

3.3 Soundness interface

The standard substitution invariant for the witnessed reconstruction is Proposition B.6; its typed Link/Comp verification is in Appendix B.

3.4 Yield-Typed Refinement and Witnesses

Definition 3.4 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and $s : \Sigma^+ \to \mathsf{S}$ a finite-semigroup typing, with canonical extension $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(\eta)}$ for $\eta \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1,\ldots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the typed nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminal* and *typed rule* refer to the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, with nonempty components if G is λ-free.

Proposition 3.5 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

The proof is given in Appendix C.

The typed homogeneity of Lemma C.3 (Appendix C) is weaker than the homogeneity of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.6 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma C.2, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \ \ (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \geq 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$. Components of canonical tuples are nonempty, so $\|\omega(X)\|_1 \geq \mathrm{fan}(X)$. Canonical objects are target witnesses, not inputs of the operator; qualitatively any choice works, while Section 4 chooses short ones.

Proposition 3.7 (Cardinality of the witness set)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be good of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and let n_{typ} and n_{rule} be the numbers of typed nonterminals and typed rules of $\widetilde{G}_0$. Then*

$$|\mathsf{W}(\widetilde{G}_0)| \leq n_{\mathrm{typ}} + n_{\mathrm{rule}} \leq |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}}.$$

Proof. One word is selected per typed nonterminal and per typed rule. A typed nonterminal is some $A_{\mathbf{t}}$ with $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, and a typed rule is determined by its source rule and the child indices $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$. □

3.5 Completeness and Exact Reconstruction

The completeness argument [14, Lemma 5] needs only that the sample exposes, for every typed rule, one same-type link and one decomposition. We state it for arbitrary typed representatives, so that it also applies to the relaxed witnesses of Section 4. No assumption that $\iota(X) \in L_X$ is made: the two required witnessed sentence words are instead explicit hypotheses.

Proposition 3.8 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\leq f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ of type $\mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ for $\widetilde{S} \to X$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1), \ldots, \iota(Y_q))]$ for each typed rule $X \to \rho(Y_1, \ldots, Y_q)$, $q \geq 0$. Then*

$$L(G) \subseteq L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) \quad (r_{\max} \geq \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed; the canonical choice $\iota = \omega$ gives $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

The verification is given in Appendix B.

Good normalization needed for exact reconstruction. A nonterminal is *reachable* if it is reachable from S in the graph with an edge $A \to B$ whenever B is a child in a rule for A, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if all nonterminals are both.

Proposition 3.9 (Good normalization with rank preservation)**.** *Let G_{in} be a finite f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$.*

The proof is given in Appendix D. The proposition controls size, fan-out and rank, not thickness; thickness transfer for reduced λ-free inputs is Proposition D.2.

Theorem 3.10 (Exact reconstruction for a fixed typing)**.** *Fix $f \geq 1$ and $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, and put the completion $\widehat{s}$. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a good presentation of $L \cap \Sigma^+$ (Proposition 3.9), and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) = L \quad \textit{for every } r_{\max} \geq \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\leq r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

The verification is given in Appendix B.

3.6 Qualitative learning and effectivity

Fixed-rank polynomial hypothesis construction (Theorem B.7) and Gold/TxtEx convergence (Corollary B.9) use Yoshinaka's conservative update policy. The proofs, the exact update-time interpretation and the unbounded-rank supplement are collected in Appendix B. These statements do not imply short characteristic data; that is the subject of Section 4.

4 Characteristic-Data Bounds

Under a rule-rank cap, Proposition 3.7 already gives polynomially many characteristic words; this section controls their total length.

The bounds by typing regime are summarized in the table in the Introduction. For arbitrary fixed typings, the refined-thickness baseline is given by Lemma 4.4.

The first row is the unconditional baseline. Full yield-type refinement can make its thickness exponentially larger than the original presentation parameters (Proposition 4.5); rank one removes the thickness dependence, and the remaining rows transfer the *original* thickness. For window-only typings the window-skeleton lemma (Lemma 4.7) gives the bound; for window–group typings the marked heavy-node lemma (Lemma 4.12) combines boundary marks and simultaneous group-product preservation in one heavy-node count, invoking the block-replacement lemma (Section 4.4.1). At fan-out one, the companion CFG paper [25, Proposition 6.4] provides a semilattice-typing obstruction to a universal original-thickness polynomial for arbitrary finite types. That result is available in arXiv version 5 and is used only for the companion-based lower-bound consequences; the positive-data upper bounds proved here do not rely on it. Polynomial update time in the observed sample is independent of these data-length refinements (Theorem B.7 and Corollary B.9).

4.1 Thickness and the witness-length template

For the quantitative results, presentations are reduced and *good*, as in Yoshinaka's fixed-dimension, fixed-rank good form [14, Section 2.3, after Example 2]. For a rule R of G, define its *rule thickness* by

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\}.$$

Thus length means the number of terminal letters in the *derived sentence*, not the number of steps of its derivation. Put $\theta_G := \max_{R \in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$, and $t_G := \max_A \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$. Thus θ_G is rule thickness and t_G nonterminal thickness; Yoshinaka writes them τ_G and t_G [14, Section 4.3]. If a target contains ε, every statement below is applied to its λ-free core and the single datum ε is added, exactly as in Theorem 3.10; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \le \theta_G$; that is, every $A \in V$ has a tuple $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.*

(iii) *$\theta_G \le 2\|G\| \max\{1, t_G\}$.*

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ and $E_A[\vec{u}] \in L(G)$ for every $\vec{u} \in L_A(G)$, and $E_S = \square_1$. For every finite typing $\widehat{s}$ and every non-start typed nonterminal $X = A_{\mathbf{t}}$ of $\widetilde{G}_0$, the choice $\chi(X) := E_A$ meets Definition 3.6, and E_A arises from a successful typed derivation by deleting an occurrence of an X-subtree.*

Proof. Fix a rule R with left-hand side A and a successful derivation D_R using R of length $\theta_G(R)$; it exists because G is reduced. By nondeletion, every terminal written by R and every component of the tuple produced at that occurrence of A survives exactly once in the final sentence. This gives (i), and also (ii) because every productive A has a rule. For (iv) with $A \neq S$, delete the subtree rooted at that occurrence of A and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root shows that the result E_A lies in $\mathcal{E}_d$, and its fixed material is part of the selected sentence. Substituting any $\vec{u} \in L_A(G)$ yields another successful derivation. Since $L_X \subseteq L_A(G)$ by Proposition 3.5(i), E_A accepts L_X; a typed nonterminal with a start link has the form $S_{\mathbf{t}}$, for which $E_S = \square_1$. Filling E_A's derivation with an X-derivation and lifting by Proposition 3.5(ii) places X at the hole, so E_A is derivational for X.

For (iii), follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Keep the path child open, complete every sibling and every child of R by a shortest terminal tuple (of length at most t_G), and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the resulting sentence has length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Hence the two thickness parameters are polynomially equivalent on reduced good presentations. Yoshinaka's efficiency criterion asks for polynomial hypothesis construction, characteristic-set cardinality, and total positive data in target size and rule thickness [14, Definition 1]; see also [10]. Proposition D.2 transfers the size–thickness scale to reduced λ-free inputs; genuine empty components may instead cause the exponential growth of Proposition D.3.

All higher-rank bounds below are assembled by the following two lemmas.

Lemma 4.2 (Witness-length template)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \ge 1$, let $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$ be a finite typing, and let $\widetilde{G}_0$ be the trimmed $\widehat{s}$-refinement. If every non-start typed nonterminal has a canonical tuple with $\|\omega(X)\|_1 \le b_{\mathrm{can}}$ and $\chi(X) := E_A$ is taken from Lemma 4.1(iv), then every word of $\mathsf{W}(\widetilde{G}_0)$ has length at most $2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}$, and*

$$|\mathsf{W}(\widetilde{G}_0)| \le |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}})(1 + 2\bar{\theta}_G + r_{\max} b_{\mathrm{can}}).$$

If moreover $L(G) = L \cap \Sigma^+$ for some $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$.

Proof. A witness is $\chi(X)[\omega(X)]$, $\chi(X)[\vec{c}]$ or $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_q))]$. The context contributes at most $\bar{\theta}_G$ terminals, the source template at most $\bar{\theta}_G$ by Lemma 4.1(i), and there are at most $r_{\max}$ child tuples. The count is Proposition 3.7; multiply. The characteristic property is Proposition 3.8 (the cap $r_{\max}$ bounds every typed rank) together with Proposition B.6 and the Start rule for ε. □

Lemma 4.3 (Type-preserving replacement)**.** *Let G be reduced and good, let $s : \Sigma^+ \to \mathsf{S}$ be a finite-semigroup typing and $\lambda : \Sigma^* \to N$ an auxiliary morphism with $s \preceq \lambda|_{\Sigma^+}$, and let $X = B_{\mathbf{t}}$ be a non-start typed nonterminal of the trimmed $\widehat{s}$-refinement $\widetilde{G}_0$. If $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(d)}(\vec{u}') = \lambda^{(d)}(\vec{u})$, then $\vec{u}' \in L_X$; in particular $\vec{u}'$ may be chosen as $\omega(X)$.*

Proof. Goodness makes all components nonempty, so $\widehat{s} \preceq \lambda$ gives $\widehat{s}^{(d)}(\vec{u}') = \mathbf{t}$. Proposition 3.5(ii) lifts a G-derivation of $\vec{u}'$ to a $G^{\widehat{s}}$-derivation rooted at X. Inserting it at an occurrence of X in a successful derivation of $\widetilde{G}_0$ gives a successful derivation, because typed rules depend only on typed symbols; hence all its rules survive trimming. □

4.2 Baselines and Rule Rank One

For a reduced good presentation G of an ε-free target, let $\widetilde{G}_0$ be its trimmed full yield-type refinement and put $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$, where $\theta_{\widetilde{G}_0}$ is rule thickness computed in the refined grammar. No comparison with the original untyped thickness is assumed.

Lemma 4.4 (Refined-thickness baseline)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$, a finite alphabet Σ, and a finite typing $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. Let G be a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of an ε-free language in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. Then the target witness set has cardinality polynomial in $\|G\|$ and total positive length polynomial in $\|G\|$ and $\bar{\theta}_{\mathrm{typ}} := \max\{1, \theta_{\widetilde{G}_0}\}$.*

Proof. $\widetilde{G}_0$ is reduced and good, so Lemma 4.1(i),(ii),(iv) applies to it with $\bar{\theta}_{\mathrm{typ}}$ in place of $\bar{\theta}_G$: canonical tuples and (derivational) canonical contexts of length at most $\bar{\theta}_{\mathrm{typ}}$ exist, with $\chi(X) = \square_1$ for start links. Each typed rule writes at most $\bar{\theta}_{\mathrm{typ}}$ terminals and has at most $r_{\max}$ children, so every witness has length at most $(r_{\max}+2)\bar{\theta}_{\mathrm{typ}}$, and

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}})(1 + (r_{\max}+2)\bar{\theta}_{\mathrm{typ}})$$

by Proposition 3.7. □

For trivial typing no productive nonterminal has more than one yield type, so the original thickness scale is retained; this recovers Yoshinaka's untyped estimates $|K_G| \leq |P|$ and $\|K_G\| \leq |P|\tau_G$ [14, Lemma 7 and Theorem 1]. The quantitative claims below therefore concern typed, multi-component settings. Without uniqueness the refinement may be much thicker:

Proposition 4.5 (Exponential blow-up of full yield-type refinement)**.** *There are a fixed two-element typing h_{c}, reduced good fan-out-one rank-two grammars G_n of size $O(n)$, and a constant bound $\theta_{G_n} \leq 2$, such that $L(G_n) = \{\mathsf{a}, \mathsf{c}\}^+$ and the trimmed full yield-type refinement $\widetilde{G}_{n,0}$ satisfies $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.*

Proof. Let $M_{\mathsf{c}} = \{1, \zeta\}$ with $\zeta^2 = \zeta$, $h_{\mathsf{c}}(\mathsf{a}) = 1$, and $h_{\mathsf{c}}(\mathsf{c}) = \zeta$. For $n \geq 1$, view

$$S \to SS \mid \mathsf{a} \mid \mathsf{c} \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to \mathsf{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathsf{c} \quad (1 \leq i \leq n)$$

as a fan-out-one MCFG. It is reduced and good, has size $O(n)$ and rule rank two, generates $\{\mathsf{a}, \mathsf{c}\}^+$, and every rule has a successful use of length at most two, so $\theta_{G_n} \leq 2$. In the full refinement, the retained copy $(A_n)_1$ can derive only words without c; recursively the unique such yield is a^{2^n}. Hence the typed rule $S_{(1)} \to (A_n)_1$ has successful thickness at least 2^n. □

This is a limitation of the full-refinement route, not of the language $\{\mathsf{a},\mathsf{c}\}^+$. Separately, the companion CFG paper [25, Proposition 6.4(iv)] proves an operator-independent exponential characteristic-data lower bound for a fixed two-element semilattice typing at fan-out one and rank two. We do not reproduce that proof here; consequences using it are explicitly attributed to the companion and do not automatically extend to higher fan-out.

Theorem 4.6 (Rank-one polynomial construction and characteristic data)**.** *Fix $f \geq 1$, finite Σ, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$. Write $\widehat{s}$ for the canonical completion and $\mathsf{S}^\sharp$ for its codomain. For any finite standard fan-out-$\leq f$, rank-≤ 1 grammar G_{in} of $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $\mathcal{B}^{\leq 1}_{f,s}$ is polynomial-time in $\|K\|_+$ and has a characteristic sample of total length $\mathrm{poly}_{f,s}(\|G_{\mathrm{in}}\|)$. The direct-start datum ε is added when $\varepsilon \in L$ without affecting this bound. The conservative learner $\mathcal{A}^{\leq 1}_{f,s}$ consequently TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f,1)$ with polynomial time per datum.*

Proof. Normalize without increasing rank (Proposition 3.9) and trim, obtaining a reduced good rank-$\leq$ 1 grammar G with $\|G\| = O_f(\|G_{\mathrm{in}}\|)$, and trim its typed refinement. Let $n_{\mathrm{typ}} \leq |V||\mathsf{S}^\sharp|^f$ be the number of non-start typed nonterminals and $b_G := \max\{1, \text{largest explicit terminal count of a rule}\} \leq \max\{1, \|G\|\}$. A terminal derivation from X is a chain of unary rules ending in a constant rule. Choose one minimizing first the yield length and then the number of rules. No typed nonterminal repeats: splicing in the later suffix would not increase the yield (nondeletion) and would shorten the chain. Hence $\|\omega(X)\|_1 \leq n_{\mathrm{typ}} b_G$. For the context, take $\Box_1$ if $\widetilde{S} \to X$ is present; otherwise follow a shortest path $\widetilde{S} \to X_1 \to \cdots \to X_p = X$ in the typed dependency graph, so $p \leq n_{\mathrm{typ}}$. Composing its unary templates and putting ordered holes for the variables of X gives, by Lemma 3.1 iterated with no siblings, a derivational context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(X)}$ with $|\chi(X)|_{\mathrm{fix}} \leq n_{\mathrm{typ}} b_G$ accepting all of L_X. Every witness word therefore has length at most $n_{\mathrm{typ}} b_G + b_G + n_{\mathrm{typ}} b_G$, and Proposition 3.7 with $r_{\max} = 1$ gives

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq (n_{\mathrm{typ}} + |P||\mathsf{S}^\sharp|^f)(1 + (2n_{\mathrm{typ}} + 1)b_G) = O_{f,s}(\|G_{\mathrm{in}}\|^3).$$

This set, with ε when $\varepsilon \in L$, is characteristic for $\mathcal{B}^{\leq 1}_{f,s}$ by Propositions 3.8 and B.6. The time and convergence claims are Theorem B.7 and Corollary B.9. □

This extends the rank-one untyped scale [14, Lemma 7] and the fan-out-one argument of [25] to arbitrary fixed finite typing at every fixed fan-out. Rank two cannot generally yield a grammar-size-only bound: the size-$O(n)$ grammar $A_0 \to \mathsf{c}$, $A_i \to A_{i-1}A_{i-1}$, $S \to A_n$ generates $\{\mathsf{c}^{2^n}\}$, which is (f,s)-tuple-substitutable for every fixed f, s (over a one-letter alphabet, sharing a sentence context forces equal total component lengths), and every characteristic sample for it contains a word of length at least 2^n.

4.3 Boundary Windows and Auxiliary Invariants

The refined thickness need not be polynomially controlled by the original good presentation (Proposition 4.5). A transfer is nevertheless possible whenever the typing is determined by a fixed prefix–suffix window together with an auxiliary finite invariant that can itself be preserved by short replacements. We use the thickness-sensitive representation model of Yoshinaka [14, Section 4.3].

Lemma 4.7 (Window skeletons)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$. Fix $k, \ell \geq 0$, a morphism $\kappa : \Sigma^* \to N$ into a finite monoid, and a finite typing $\widehat{s} \preceq (\widehat{s}_{k,\ell}, \kappa)$. Let $\widetilde{G}_0$ be the trimmed $\widehat{s}$-refinement and n_{typ} its number of non-start typed nonterminals. Suppose that $\beta \geq 0$ is such that for every $C \in V$ and $\vec{x} \in L_C(G)$*

there is $\vec{x}' \in L_C(G)$ *with* $\kappa^{(\mathrm{fan}(C))}(\vec{x}') = \kappa^{(\mathrm{fan}(C))}(\vec{x})$ *and* $\|\vec{x}'\|_1 \le \beta$. *Then every non-start typed nonterminal* X *has a canonical tuple with*

$$\|\omega(X)\|_1 \le f(k+\ell+1)\, n_{\mathrm{typ}}\, (\bar{\theta}_G + r_{\max}\beta).$$

For trivial κ, $\beta = \bar{\theta}_G$ *is admissible by Lemma 4.1(ii), so* $\|\omega(X)\|_1 \le f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$ *whenever* $\widehat{s} \preceq \widehat{s}_{k,\ell}$.

Proof. Choose a terminal derivation D from X in $\widetilde{G}_0$ with the minimum number of grammar-rule nodes, and let $\vec{u}$ be its root tuple. No typed nonterminal occurs twice on a root-to-leaf path: replacing the upper subtree by the lower one would give a derivation from X with fewer nodes. Hence every path has at most n_{typ} grammar nodes.

Mark in each component of $\vec{u}$ the terminals that determine its $\widehat{s}_{k,\ell}$-type (Definition 2.4): all terminals of a component shorter than $m_0 = \max\{1, k+\ell\}$, and otherwise its first k and last ℓ terminals (one arbitrary terminal when $k = \ell = 0$). There are at most $f(k+\ell+1)$ marks. Each terminal is written explicitly by exactly one node, so the union $\mathcal{K}$ of the root paths to the marked terminals has at most $f(k+\ell+1)n_{\mathrm{typ}}$ nodes. Call a child of a node of $\mathcal{K}$ that is not itself in $\mathcal{K}$ *omitted*. By linearity and nondeletion, each component of an omitted child is one contiguous block of exactly one root component, and this block contains no mark.

Replace each omitted subtree, with underlying nonterminal C and yield $\vec{x}$, by a G-derivation of a tuple $\vec{x}'$ as in the hypothesis, and forget all typed labels; this gives a G-derivation from the underlying nonterminal B of X with root tuple $\vec{u}'$. A short root component consists only of marked terminals, so it receives no replaced block and is unchanged. In a long component every replaced block lies strictly between the marked first k and last ℓ terminals, and goodness keeps it nonempty; the marked prefix and suffix remain literal, and the component stays on the long side of the threshold. Hence $\widehat{s}_{k,\ell}$-vectors agree. Each root component is the ordered product of explicit terminals and child components, and every replaced block keeps its κ-value; so κ-vectors agree, with no commutativity used. Thus $(\widehat{s}_{k,\ell}, \kappa)^{(d)}(\vec{u}') = (\widehat{s}_{k,\ell}, \kappa)^{(d)}(\vec{u})$, and Lemma 4.3 gives $\vec{u}' \in L_X$. Each node of $\mathcal{K}$ writes at most $\bar{\theta}_G$ terminals (Lemma 4.1(i)) and has at most $r_{\max}$ omitted children, each of length at most β; the bound follows. □

Theorem 4.8 (Polynomial data for prefix–suffix-determined typings)**.** *Fix* $f \ge 1$, $r_{\max} \ge 1$, k, ℓ, *a finite alphabet* Σ, *and a finite-semigroup typing* $s : \Sigma^+ \to \mathsf{S}$ *satisfying* $s \preceq s_{k,\ell}$; *write* $\widehat{s}$ *for its canonical completion. Let* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ *and let* G *be a reduced good fan-out-*$\le f$, *rank-*$\le r_{\max}$ *grammar with* $L(G) = L \cap \Sigma^+$. *Then* $\mathcal{B}^{\le r_{\max}}_{f,s}$ *is polynomial-time in the positive sample. A characteristic sample for* L *is* $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$, *with*

$$|\mathsf{W}(\widetilde{G}_0)| = O_{f,r_{\max},s}(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,s,\Sigma}(\|G\|^2\bar{\theta}_G),$$

where $\bar{\theta}_G = \max\{1, \theta_G\}$. *For a reduced* λ*-free input presentation* G_{in} *(deleting rules allowed), the structural good-form conversion of Proposition D.2 yields the explicit presentation-relative bound*

$$O_{f,r_{\max},k,\ell,s,\Sigma}\Big(\|G_{\mathrm{in}}\|^3 \max\{1, t_{G_{\mathrm{in}}}\}\Big).$$

Proof. By Lemma C.1 we may use $\widehat{s}$. Lemma 4.7 with trivial κ gives $b_{\mathrm{can}} = f(k+\ell+1)n_{\mathrm{typ}}(r_{\max}+1)\bar{\theta}_G$, and $n_{\mathrm{typ}} \le |V||\mathsf{S}^\sharp|^f = O(\|G\|)$. Lemma 4.2 bounds every witness by $O(\|G\|\bar{\theta}_G)$ and their number by $O(\|G\|)$, and gives the characteristic property; the datum ε adds at most one. Theorem B.7 gives the construction time. For the input clause, Proposition D.2 gives $\|G\| = O(\|G_{\mathrm{in}}\|)$ and $\bar{\theta}_G = O(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\})$; nonterminal thickness is used because deleting source rules may erase material before good-form conversion. □

Corollary B.9 gives polynomial time per datum for the capped conservative learner. At fan-out one, polynomial time-and-thick-data learning for fixed windows is already due to Yoshinaka [12, Theorem 1]. As explained in the Introduction, our quadratic total-encoded-length bound is measured relative to a reduced good presentation; the degree in that presentation size is uniform for fixed window parameters, without claiming an improvement over Yoshinaka's bounds after normalization. The contribution of Lemma 4.7 is to preserve boundary information simultaneously across *multiple* tuple components at every fixed $f \geq 2$. For fan-out-one window–group sufficient conditions, we credit the companion CFG manuscript [25]. Theorem 5.3 shows that this higher-rank regime is needed at the language level: it gives a non-context-free $(1,1)$-typed target of minimum fan-out two and minimum rule rank two over all finite fan-outs, whereas the rank-one examples are covered by Theorem 4.6.

4.4 Finite-Group Invariants

Window–group typings require short tuples with *prescribed* componentwise group values. The length bound is obtained from a single count of heavy nodes, which we state for an arbitrary finite-monoid invariant because Section 4.5.1 reuses it.

Lemma 4.9 (Heavy-node count)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, let $\lambda : \Sigma^* \to N$ be a morphism into a finite monoid, and let $b \geq \bar{\theta}_G$ and $c \geq 0$. Let T be a terminal derivation from $B \in V$ that minimizes first the total yield length and then the number of nodes among derivations from B with the same root λ-vector. Call a node* heavy *if its subtree yield has total length greater than b. If every set of pairwise incomparable heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \leq (r_{\max}+1)\, b\, \max\{1, c\,|V||N|^f\}.$$

Proof. Label each node by its nonterminal and the λ-vector of its yield; there are at most $|V||N|^f$ labels. If an ancestor and a proper descendant had the same label, replacing the former subtree by the latter would keep the root λ-vector (templates evaluate homomorphically, as in Proposition 3.5), would not increase the yield (nondeletion) and would remove nodes. So every root-to-leaf path has at most $|V||N|^f$ nodes. If the root is light, its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree, whose leaves are pairwise incomparable; hence there are at most $c|V||N|^f$ heavy nodes. Every terminal of the root yield is written either by a heavy node (at most $\bar{\theta}_G \leq b$ per node, Lemma 4.1(i)) or inside a light child of a heavy node (at most $r_{\max}$ children of length at most b each). □

For a finite abelian group H write $\mathsf{d}(H)$ for the maximal length of a zero-sum-free sequence over H, so that $\mathsf{D}(H) = \mathsf{d}(H) + 1$ is its Davenport constant. For a finite group A and $f \geq 1$ put

$$N_{A,f} = \max\{3, |A|^f\}, \quad U_f = f^{f+1} f!, \quad M_{A,f} = U_f\, N_{A,f}^{2^{f^2}}, \qquad c_{A,f} := \begin{cases} \mathsf{d}(A^f), & A \text{ abelian}, \\ M_{A,f} - 1, & \text{otherwise}. \end{cases}$$

The lower floor $N_{A,f} \geq 3$ is required for the mixed-direction cases in the block-order decomposition; the $|A|^f$ term supplies repeated group boundary states.

Proposition 4.10 (Simultaneous block replacement: the exact interface)**.** *Let A be a finite group and $f \geq 1$. There are d ordered output rows ($1 \leq d \leq f$). Row c is a fixed-order product of q_c blocks, interspersed with unaltered constants $a_{c,0}, \ldots, a_{c,q_c} \in A$. A block $\beta_{c,j}$ has an owner $o(c,j) \in [m]$ and an original and a replacement value $x_{c,j}, x'_{c,j} \in A$. Every owner has at most f blocks over all rows.*

Different owners' blocks may interleave arbitrarily; all material not replaced is absorbed into the intervening constants. For $J \subseteq [m]$ put

$$\mathrm{val}_c(J) = a_{c,0} y_{c,1} a_{c,1} \cdots y_{c,q_c} a_{c,q_c}, \qquad y_{c,j} = \begin{cases} x'_{c,j}, & o(c,j) \in J, \\ x_{c,j}, & o(c,j) \notin J. \end{cases}$$

With $c_{A,f}$ as defined above, if $m > c_{A,f}$ there exists a nonempty $J \subseteq [m]$ such that $\mathrm{val}_c(J) = \mathrm{val}_c(\varnothing) \quad (c = 1, \ldots, d)$. For abelian A, one may take $c_{A,f} = \mathsf{d}(A^f)$; for an arbitrary finite group, $c_{A,f} = M_{A,f} - 1$ suffices. Individual block or owner values need not be preserved.

Proof. This is Lemma 4.17 of Section 4.4.1, written with root-component notation. For abelian A, associate with each owner its difference vector in A^d and select a nonempty zero-sum subsequence. In a nonabelian group one must not commute factors. Instead fix a common owner pattern and homogenize pairwise order comparisons by Lemma 4.21; Lemma 4.19 then decomposes the output rows into consecutive rank chunks, and Lemma 4.20 selects a nonempty interval of owners preserving all ordered row products. The threshold is at most $M_{A,f}$. The detailed extraction and constants appear in Section 4.4.1. □

Theorem 4.11 (Short tuples with prescribed finite-group values)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and maximum rule rank $r_{\max} \geq 1$. Let $g : \Sigma^* \to A$ be a homomorphism into* any *finite group A. For every $B \in V$ and every realizable $\mathbf{s} \in A^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $g^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{s}$ and*

$$\|\vec{u}\|_1 \leq \beta_g := (r_{\max} + 1)\bar{\theta}_G \max\{1, c_{A,f}|V||A|^f\}.$$

In particular $\beta_g = O_{A,f,r_{\max}}(\|G\|\bar{\theta}_G)$; when A is abelian the antichain constant is the zero-sum constant $\mathsf{d}(A^f)$.

Proof. Let T be minimal as in Lemma 4.9 with $\lambda = g$ and $b = \bar{\theta}_G$, and let $v_1, \ldots, v_m$ be pairwise incomparable heavy nodes. For each v_i, labelled C_i, fix a C_i-derivation of length at most $\bar{\theta}_G$ (Lemma 4.1(ii)); for $J \subseteq [m]$ let T_J replace the subtrees at v_i, $i \in J$. By linearity and nondeletion each component of v_i is a contiguous block of one root component, and blocks of different v_i are disjoint. The g-value of a root component of T_J is the ordered product of the g-values of its fixed segments and of its blocks, each block contributing its original or its replacement value according to whether its owner lies outside or inside J. This is a block-replacement family over A in the sense of Definition 4.16. If $m > c_{A,f}$, Proposition 4.10 yields a nonempty J for which T_J has the root g-vector of T; but T_J is strictly shorter, since every replaced subtree has length greater than $\bar{\theta}_G$. This contradicts minimality, so Lemma 4.9 applies with $c = c_{A,f}$ and $N = A$. □

Lemma 4.12 (Marked heavy nodes for simultaneous window–group shortening)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$. Fix $k, \ell \geq 0$, a finite group A, and a homomorphism $g : \Sigma^* \to A$. Put $\lambda = (\widehat{s}_{k,\ell}, g) : \Sigma^* \to N := M_{k,\ell} \times A$, $h := f(k + \ell + 1)$, and $n_\lambda := |V||N|^f$. For every $B \in V$ and realizable vector $\mathbf{t} \in N^{\mathrm{fan}(B)}$, there exists $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq b_{\mathrm{mark}} := (r_{\max} + 1)\bar{\theta}_G \max\{1, (c_{A,f} + h)n_\lambda\}.$$

Hence $b_{\mathrm{mark}} = O(\|G\|\bar{\theta}_G)$ with $f, r_{\max}, k, \ell, A, g, \Sigma$ fixed. If A is abelian, $c_{A,f}$ can be taken as $\mathsf{d}(A^f)$.

Proof. Among B-derivations with root componentwise λ-value $\mathbf{t}$, choose T minimizing total yield length, then node count. As in Lemma 4.9, a root–leaf path repeats no pair of a nonterminal and its componentwise λ-value; its number of nodes is at most n_λ.

Mark the first k and last ℓ terminal positions in each root component of length at least $m_0 := \max\{1, k+\ell\}$, and all positions in a shorter component. If $k = \ell = 0$, mark one position per component to ensure that nonemptiness is explicitly preserved. There are at most $h = f(k+\ell+1)$ marks in total.

Call a node heavy if its subtree yields more than $\bar{\theta}_G$ terminals. Fix an arbitrary antichain Q of heavy nodes. Since incomparable subtrees occupy disjoint sets of terminal positions, at most h nodes of Q contain marked terminals. Let Q_0 consist of all other nodes. For each $v \in Q_0$ with nonterminal C_v, fix a derivation producing a tuple of total length at most $\bar{\theta}_G$, which exists by Lemma 4.1(ii). Each replacement strictly shortens the original heavy subtree.

Suppose $|Q_0| > c_{A,f}$. By linearity and nondeletion, every node in Q_0 contributes at most f nonoverlapping blocks distributed among at most f ordered root components. Everything outside these subtrees remains fixed. Since g is a homomorphism, each root group value is the ordered product of the unchanged pieces and the original or replacement block values. Proposition 4.10 thus selects a nonempty set of owners whose simultaneous replacement preserves all root group values, without assuming commutativity.

No selected subtree contains a mark. Short root components are untouched, while the marked prefixes and suffixes of long components retain their letters and order. Every replacement component is nonempty by goodness. Consequently all root $\widehat{s}_{k,\ell}$-values are preserved. The root λ-vector is unchanged, while the total yield length strictly decreases, a contradiction to the choice of T. Every heavy antichain therefore has at most $c_{A,f} + h$ members. Lemma 4.9, with $b = \bar{\theta}_G$, $c = c_{A,f} + h$ and $N = M_{k,\ell} \times A$, gives the displayed bound. For abelian A use Lemma 4.17(a). □

Example 4.13 (Two components with crossed owner order). Two owners with two blocks each may appear in the root tuple in the order $(\beta_{1,1}\beta_{2,1},\ \beta_{2,2}\beta_{1,2})$. Take $A = S_3$, put $\sigma = (12)$ and $\tau = (23)$, and assign value σ to the blocks of owner 1 and τ to those of owner 2. The root group vector is $(\sigma\tau, \tau\sigma)$, with $\sigma\tau \neq \tau\sigma$. Thus commuting owner contributions would be invalid. This illustrates a block-replacement arrangement, not a proof that any particular target language requires a nonabelian typing. Proposition 4.10 preserves the two ordered root products simultaneously.

Theorem 4.14 (Polynomial characteristic data for window–finite-group typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite group A, a homomorphism $g : \Sigma^* \to A$, and a fixed positive-word typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ whose fresh-identity completion satisfies $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$. For every target $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped reconstruction operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,\widehat{s},g,\Sigma}(\|G\|^2 \bar{\theta}_G).$$

Proof. Apply Lemma 4.12 to $\lambda = (\widehat{s}_{k,\ell}, g)$. Since $\widehat{s} \preceq \lambda$, every realizable $\widehat{s}$-typed non-start nonterminal $X = B_{\mathbf{s}}$ has an untyped B-tuple of the same root λ-value and total length at most $b_{\mathrm{mark}} = O(\|G\|\bar{\theta}_G)$. Lemma 4.3 lifts its derivation to X, so $b_{\mathrm{can}} \leq b_{\mathrm{mark}}$. The witness template (Lemma 4.2) gives $O(\|G\|)$ witnesses, each of length $O(\|G\|\bar{\theta}_G)$; the total encoded length is $O(\|G\|^2\bar{\theta}_G)$. Handle ε by its direct start rule. □

Theorem 4.15 (Polynomial data for window–abelian typing)**.** *Fix $f, r_{\max}, k, \ell, \Sigma$, a finite abelian group A, a homomorphism $g : \Sigma^* \to A$, and a fixed positive-word typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ whose fresh-identity completion satisfies $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$,*

rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped operator $\mathcal{B}_{f,s}^{\leq r_{\max}}$ has a characteristic sample satisfying

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},k,\ell,\widehat{s},g,\Sigma}(\|G\|^2 \bar{\theta}_G).$$

In particular the bound uses the marked-node constant

$$b_{\mathrm{mark}} = (r_{\max}+1)\bar{\theta}_G \max\{1, (\mathsf{d}(A^f) + f(k+\ell+1))|V|(|M_{k,\ell}||A|)^f\}.$$

Proof. Apply the proof of Theorem 4.14 using the abelian case of Lemma 4.12. Lemma 4.17(a) permits $c_{A,f} = \mathsf{d}(A^f)$ rather than $M_{A,f} - 1$, with $\mathsf{d}(A^f) = O(\exp(A) f \log|A|)$ (Remark 4.18). □

Commutativity is used only in the zero-sum case of Lemma 4.17. For nonabelian A, the order-homogenization proof of Section 4.4.1 controls blocks interleaving across multiple root components.

4.4.1 Proof of simultaneous block replacement in finite groups

Semigroup convention. The finite group A and the morphisms into it are auxiliary to the shortening proofs. The semantic typing is always the positive-image restriction of the window–group (or local-group) product, with the fresh identity used only to evaluate empty template factors; by Lemma C.1 this changes neither nonempty tuple comparisons nor the selected witnesses.

We now prove the purely group-theoretic statement used in Theorem 4.11 and Lemma 4.30. The commutative case is a zero-sum argument; the general case needs an order-homogenization argument because, for $f \geq 2$, blocks of different indices can interleave differently on different lines. The proof has three logically separate parts: the classification of homogeneous point orders (Lemma 4.19), the group-valued interval invariant (Lemma 4.20), and the polynomial-size extraction of homogeneous indices (Lemma 4.21). Neither the order classification nor the selection of a replacement interval assumes that the group is abelian.

Definition 4.16 (Block-replacement family)**.** Let A be a finite group and $f \geq 1$. A *block-replacement family* of width f over A with index set $[m]$ consists of $d \leq f$ *lines*; on line c, a left-to-right sequence of blocks $\beta_{c,1}, \dots, \beta_{c,q_c}$ and constants $a_{c,0}, \dots, a_{c,q_c} \in A$; for every block β an *owner* $o(\beta) \in [m]$ and two values $x_\beta, x'_\beta \in A$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_c(J) := a_{c,0}\, y_{c,1}\, a_{c,1} \cdots y_{c,q_c}\, a_{c,q_c}, \qquad y_{c,j} := \begin{cases} x'_{\beta_{c,j}}, & o(\beta_{c,j}) \in J, \\ x_{\beta_{c,j}}, & o(\beta_{c,j}) \notin J. \end{cases}$$

Lemma 4.17 (Block replacement)**.** *In a block-replacement family of width f over A there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ for every line c, provided that either* (a) *A is abelian and $m > \mathsf{d}(A^f)$, or* (b) *A is arbitrary and $m \geq M_{A,f}$, where d and $M_{A,f}$ are as in Section 4.4.*

Part (a) is proved below under commutativity, and part (b) under the arbitrary-group order homogenization argument.

Commutative groups: zero sums

Proof of Lemma 4.17(a). Write A additively. By commutativity $\mathrm{val}_c(J) - \mathrm{val}_c(\emptyset) = \sum_{i \in J} \delta_{i,c}$, where $\delta_{i,c}$ is the sum of $x'_\beta - x_\beta$ over the blocks β of line c owned by i. Put $\delta_i = (\delta_{i,c})_c \in A^d$. A sequence over A^d of length $m > \mathsf{d}(A^f) \geq \mathsf{d}(A^d)$ has a nonempty zero-sum subsequence $(\delta_i)_{i \in J}$; the inequality holds because A^d embeds in A^f and a zero-sum-free sequence in a subgroup is zero-sum free in the group. □

Remark 4.18 (Size of the zero-sum constant). The prefix-sum pigeonhole argument gives $\mathsf{d}(A^f) \le |A|^f - 1$; the zero-sum bound is usually much stronger. Write $A \cong C_{n_1} \oplus \cdots \oplus C_{n_s}$ and $n = \exp(A) \ge 2$. Then

$$f \sum_{i=1}^{s} (n_i - 1) \;\le\; \mathsf{d}(A^f) \;\le\; (n-1) + n \log \frac{|A|^f}{n},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [33, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [33, Theorem 5.5.9] (Olson). Hence $\mathsf{d}(A^f) = O(\exp(A)\, f \log |A|)$ is linear in f, whereas $|A|^f - 1$ is exponential; for $A = C_2$ the antichain bound drops from $2^f - 1$ to f. The value $\mathsf{d}(A^d)$ is also the limit of the argument as stated, since a zero-sum-free sequence of that length exists; we do not claim that a grammar realizes such a configuration.

Arbitrary finite groups: order homogenization

Lemma 4.19 (A homogeneous block-order lemma)**.** *Fix positive integers e, n with $n \ge 3$. For each $t \in [e]$ and $i \in [n]$, let (t, i) be a distinct point on a line, with a total left-to-right order. Suppose that (i) the order of (t, i) and (u, i) is independent of i; and (ii) for each ordered pair (t, u), including $t = u$, the order of (t, i) and (u, j) for $i < j$ is independent of the choice of $i < j$. Then the types at the points have a partition into classes, each occupying a consecutive region of the line. Every class has one direction (increasing or decreasing in i); inside its region the points occur in n successive* chunks*, one per rank in that direction. Each chunk consists of exactly one point of each type in the class (in a fixed within-chunk order).*

Proof. By (ii) with $t = u$, each type's points are monotone. For distinct types t, u write X for $(t, i) < (u, j)$ and Y for $(u, i) < (t, j)$, $i < j$; both are uniform by (ii). Suppose first that both types increase. $X \wedge Y$ puts all points of rank i before all points of rank $j > i$. Under $X \wedge \neg Y$ all off-diagonal comparisons put t left of u, and so do equal ranks: $(u, i) < (t, i)$ would give $(t, 1) < (t, 2) < (u, 1) < (t, 1)$; thus the types are separated. The case $\neg X \wedge Y$ is symmetric, and $\neg X \wedge \neg Y$ gives $(t, 1) < (t, 2) < (u, 1) < (u, 2) < (t, 1)$, a contradiction. If t increases and u decreases, $X \wedge Y$ gives $(t, 2) < (u, 3) < (u, 1) < (t, 2)$ and $\neg X \wedge \neg Y$ gives $(t, 3) < (u, 2) < (t, 1) < (t, 3)$; under $X \wedge \neg Y$ all off-diagonal comparisons put t left of u, and a contrary equal-rank comparison would give $(t, 1) < (u, 2) < (u, 1) < (t, 1)$, contrary to (i), so the types are separated; $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types. Thus any two types are either *separated*, or have the same direction and are *interleaved by rank*.

Interleaving is transitive: if t interleaves with u and u with v, all three have one direction, say increasing; if t, v were separated with t to the left we would have $(v, 1) < (u, 2) < (t, 3) < (v, 1)$, and with v to the left $(t, 1) < (u, 2) < (v, 3) < (t, 1)$. Hence interleaving, with equality, partitions the types into classes. A type outside a class is separated from each type inside it and lies on the same side of all of them, since otherwise two interleaved types would be separated. So class regions are consecutive, the rank comparisons give successive chunks, and the within-chunk order is independent of rank by (i). □

Lemma 4.20 (Simultaneous finite-group interval shortening)**.** *Let A be any finite group, and consider a block-replacement family of width f over A with $n \ge \max\{3, |A|^f\}$ indices in which all indices have the same number of blocks, the same lines for them and the same within-index order, and in which every cross-index comparison between the t-th block of index i and the u-th block of index j on a common line has the same truth value for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_c(J) = \mathrm{val}_c(\emptyset)$ for every line c.*

Proof. Apply Lemma 4.19 on each line to the block types going to that line. Across all lines there are $q \le f$ class regions. For such a region K, enumerate its chunks $1, \dots, n$ from left to right, including the constants between the blocks of the region. Let P_p be the original value of its prefix ending after chunk p and Q_p the value of that prefix when chunks $1, \dots, p$ are replaced; set $P_0 = Q_0 = 1_A$ and $\kappa_p^K = P_p Q_p^{-1}$. For $0 \le p < p' \le n$, the *ordered* value after replacing exactly chunks $p+1, \dots, p'$ is

$$\underbrace{P_p}_{\text{old prefix}} \underbrace{(Q_p^{-1} Q_{p'})}_{\text{new middle}} \underbrace{(P_{p'}^{-1} P_n)}_{\text{old suffix}}.$$

The middle expression is correct even with constants between chunks: $Q_{p'} = Q_p$ (new middle) by the definition of prefix products. Therefore the displayed product equals P_n exactly when

$$P_p Q_p^{-1} Q_{p'} P_{p'}^{-1} = 1_A, \quad \text{equivalently} \quad \kappa_p^K = P_p Q_p^{-1} = P_{p'} Q_{p'}^{-1} = \kappa_{p'}^K.$$

No factors have been commuted. Note that this requires equality of *two boundary states*; it does not require an individual block's original and replacement values to agree.

If K increases with the index order, replacing owners $(a, b]$ replaces chunks $(a, b]$ in the region. If the order decreases, the same owners occupy chunks $(n-b, n-a]$ (counted from the left). This reversal is why the two orientations must use distinct state indices. For $j = 0, \dots, n$ define a state in A^q by taking κ_j^K from each increasing and κ_{n-j}^K from each decreasing class. There are $n+1 > |A|^q$ states, so two coincide, at $a < b$. Replacing the indices $a+1, \dots, b$ then preserves every region value; everything else on every line is unchanged. □

Independent Ramsey proof of the qualitative assertion in Lemma 4.17(b). We first establish existence of a simultaneous replacement set without using the quantitative order-homogenization bound below. Let $N = \max\{3, |A|^f\}$, and let $R(N, c)$ denote a finite two-uniform Ramsey number for c colors. An owner with no blocks can be selected alone. Otherwise, among sufficiently many owners, select $R(N, 2^{f^2})$ of the same unary pattern (number of blocks, assigned lines, and within-owner order); there are at most U_f patterns. Absorb all other owners into the unchanged constants.

The common pattern has $e \le f$ block types. Color a pair of selected owners $i < j$ by the truth values of $p_t(i) < p_u(j)$ for all ordered pairs of types (t, u) on the same line. There are at most $2^{e^2} \le 2^{f^2}$ colors. By finite Ramsey, there is an ordered set of N owners on which every comparison has a fixed truth value, independently of $i < j$. Within-owner comparisons are already fixed by the common pattern. On each line, the argument of Lemma 4.19 decomposes this homogeneous configuration into consecutive rank-ordered regions, all increasing or decreasing. Lemma 4.20 now supplies a nonempty interval of owners whose simultaneous replacement leaves the ordered product on every line unchanged.

This proves the *existence* assertion for an arbitrary finite, possibly nonabelian group, without assuming any commutation of factors. Ramsey's bound does not supply the asserted polynomial dependence on $|A|$ for fixed f; that stronger, explicit assertion is established next by the monotone-comparison extraction. □

Lemma 4.21 (Polynomial order homogenization)**.** *Let $N \ge 3$, and define $m_0 = N$ and $m_{s+1} = (m_s - 1)^2 + 1$, so that $m_k \le N^{2^k}$. Let T be a finite set of point types, each assigned to one of several lines. For every index $i \in [n]$ and $t \in T$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let Π be the set of ordered pairs (t, u), including $t = u$, of types on a common line, and put $k = |\Pi|$. If $n \ge m_k$, there is a set $I \subseteq [n]$ of N indices such that, for every $(t, u) \in \Pi$, the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t,t) first. Each step passes from a current index set of size at least $(m-1)^2+1$ to a subset of size at least m; homogeneity already obtained, and monotonicity of each type, persist in subsets. For (t,t), homogeneity means that p_t is monotone, and the Erdős–Szekeres theorem gives a monotone subsequence of length m.

For $t \neq u$, both p_t and p_u are already monotone. Relabel the current indices $1,\dots,m'$ in order, with $m' \geq (m-1)^2+1$, and put $X(i,j) = [p_t(i) < p_u(j)]$ for $i<j$. Because $p_u(j)$ is monotone in j, every row $X(i,\cdot)$ is monotone; because $p_t(i)$ is monotone in i, every column $X(\cdot,j)$ is monotone. The respective directions are fixed over all rows and columns by the directions of the two point types. Replacing X by $1-X$ if necessary, assume each row goes from 0 to 1.

If the columns go from 1 to 0, put $J(i) = \min\{j > i : X(i,j) = 1\}$ (or ∞); J is nondecreasing. Set $i_1 = 1$ and $i_{s+1} = J(i_s)$ while this is at most m'. For $a<b$, $i_b \geq J(i_a)$ and so $X(i_a,i_b) = 1$; if the chain has m terms we are done. Otherwise its at most $m-1$ terms cut $[m']$ into at most $m-1$ intervals $[i_s, i_{s+1})$ and $[i_L, m']$, one of which has more than $m'/(m-1) > m-1$ indices; inside it every $i<j$ satisfies $j < J(i_s) \leq J(i)$, so $X(i,j) = 0$ (and $J \equiv \infty$ in the last interval).

If the columns go from 0 to 1, the set $\{X=1\}$ is closed upward in both coordinates. Let κ be the least index with $X(\kappa,\kappa+1) = 1$, or $\kappa = m'$. For $\kappa \leq i < j$, $X(i,j) \geq X(\kappa,\kappa+1) = 1$; for $i<j\leq\kappa$, $X(i,j) \leq X(j-1,j) = 0$. One of $[1,\kappa]$ and $[\kappa,m']$ has at least $(m'+1)/2 \geq m$ indices, because $(m-1)^2+1 \geq 2m-1$.

After all k pairs at least $m_0 = N$ indices remain, and $m_{s+1} \leq m_s^2$ gives $m_k \leq N^{2^k}$. □

Proof of Lemma 4.17(b). If an index owns no blocks, choosing that index alone preserves every line, so assume each index owns at least one block. The *unary pattern* of an index records the number of its blocks, the line of each, and their order within any shared line; there are at most $f \cdot f^f \cdot f! = U_f$ patterns. Hence at least $N_{A,f}^{2^{f^2}}$ indices share one pattern; the blocks of all other indices are never replaced and are absorbed into the constants. Number the blocks of the common pattern as types $1,\dots,f'$ ($f' \leq f$) and use their positions as points on their lines. There are at most f^2 ordered pairs of types on a common line, including $t=u$, and $m_k \leq N^{2^k} \leq N^{2^{f^2}}$ for $k \leq f^2$, so Lemma 4.21 with $N = N_{A,f}$ yields $N_{A,f}$ indices with constant cross-index comparisons for all $i<j$. These satisfy the hypotheses of Lemma 4.20, which supplies the required nonempty interval. □

For fixed f the constant $M_{A,f} = U_f \max\{3,|A|^f\}^{2^{f^2}}$ is polynomial in $|A|$; it replaces the multicolor Ramsey bound $U_f\, c_f^{\,c_f(N_{A,f}-1)+1}$, $c_f = 2^{f^2}$, of earlier versions, which is exponential in $|A|^f$. The dependence on f remains doubly exponential.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

Write **LI** for the pseudovariety of finite locally trivial semigroups and **Grp** for the pseudovariety of finite groups. Their join $\mathbf{LI} \vee \mathbf{Grp}$ consists of divisors of finite products $S \times A$, with $S \in \mathbf{LI}$ and A a finite group [27, 28]. The following criterion makes the window–group hypothesis intrinsic to the positive image of the fixed typing; its finite-semigroup ingredient also occurs in the companion CFG manuscript [25].

Lemma 4.22 (Locally trivial–group typings admit window–group refinements)**.** *Let Σ be a nonempty finite alphabet and let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be a finite-semigroup typing; write $\widehat{s}$ for its canonical completion. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, then there exist $k,\ell \geq 0$, a finite group A, and a morphism $g : \Sigma^* \to A$ such that $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ on all of Σ^*. If $\mathsf{S} \in \mathbf{LI}$, the group factor can be taken trivial, so $\widehat{s} \preceq \widehat{s}_{k,\ell}$.*

Proof. Choose a finite locally trivial semigroup S, a finite group A, a subsemigroup $T \subseteq S \times A$, and an onto semigroup morphism $\pi : T \to \mathsf{S}$. Lifting each $s(a)$ to some $(s_a, g_a) \in T$ defines a morphism

$\rho : \Sigma^+ \to T$ with $\pi\rho = s$; let $\sigma : \Sigma^+ \to S$ and $g : \Sigma^* \to A$ (with $g(\varepsilon) = 1_A$) be its coordinates. The positive image $S_0 = \sigma(\Sigma^+)$ is locally trivial. Put $q = |S_0| + 1$. The finite locally trivial identities $S_0^q = S_0 E(S_0) S_0$ and $euf = ef$ for idempotents e, f and $u \in S_0$ [26] show that $\sigma(w)$ is determined by the length-q prefix and suffix of w when $|w| \geq 2q$: writing $w = pxz$ with $|p| = |z| = q$, $\sigma(p) = aeb$, $\sigma(z) = cfd$ and $\sigma^1(\varepsilon) = 1$, $\sigma(pxz) = ae\,(b\,\sigma^1(x)\,c)\,fd = aefd = \sigma(pz)$, where the parenthesized factor lies in S_0 because $b, c \in S_0$. Words shorter than $2q$ are recorded exactly by $\widehat{s}_{q,q}$. Hence equal $(\widehat{s}_{q,q}, g)$-values on Σ^+ give equal ρ- and $\widehat{s}$-values; the window typing separates ε from Σ^+. Take $k = \ell = q$. If $\mathsf{S} \in \mathbf{LI}$, use $S = \mathsf{S}$, $\pi = \mathrm{id}$ and the trivial group. □

Corollary 4.23 (Algebraic sufficient conditions for polynomial data)**.** *Fix a nonempty finite Σ, a fan-out cap $f \geq 1$, a rule-rank cap $r_{\max} \geq 1$, and a finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$; write $\widehat{s}$ for its canonical completion. Let G be any reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of a target in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, and put $\bar{\theta}_G = \max\{1, \theta_G\}$. The capped reconstruction operator has a characteristic sample of cardinality $O_{f,r_{\max},s}(\|G\|)$ with:*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G) \qquad \text{if } \mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}.$$

The locally trivial case **LI** *is included, with a trivial group factor.*

Proof. Combine Lemma 4.22 with Theorem 4.8 or Theorem 4.14; the refining parameters depend only on s and Σ, not on G. □

The local-group characterization and shortening proof are given below in Section 4.5.1. A finite semigroup belongs to **LG** (the *local groups*) if eSe is a group for every idempotent $e \in S$. This condition concerns the *positive image*, never the fresh-identity completion.

Theorem 4.24 (Polynomial characteristic data for local-group typing)**.** *Fix $f, r_{\max}, \Sigma$ and a finite-semigroup typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{LG}$; write $\widehat{s}$ for its canonical fresh-identity completion. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G with $L(G) = L \cap \Sigma^+$, the capped operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^3 \bar{\theta}_G).$$

Proof roadmap. The local-group representation of Theorem 4.28 refines $\widehat{s}$ by a fixed prefix–suffix window and an ordered product of finite-group values attached to sliding factors. The window skeleton gives short replacement candidates of length $O(\|G\|\bar{\theta}_G)$. A simultaneous noncommutative block-replacement argument chooses which candidates to use while preserving every root component's ordered group product, yielding typed canonical tuples of length $O(\|G\|^2\bar{\theta}_G)$. The witness-length template counts $O(\|G\|)$ typed rules. The proof is given in Section 4.5.1 below.

The inclusion $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$ is strict (Proposition 4.29). Thus the local-group bound already strictly strengthens the companion CFG sufficient regime when $f = 1$, while the algebraic characterization of local groups itself is classical. At fan-out one the two-element semilattice supplies the pseudovariety-level lower-bound obstruction (Corollary 4.31); that corollary's negative direction uses the publicly available companion CFG result [25, Proposition 6.4]. No corresponding lower bound at fan-out $f \geq 2$ is asserted; see Open Problem 6.4. The empty-alphabet case is trivial.

4.5.1 Local-group sliding-factor representation and shortening

Semigroup convention. Throughout this subsection **LG** is tested on the *positive image* $S = \widehat{s}(\Sigma^+)$ of the typing, not on the canonical completion $S^\sharp$. Auxiliary group-valued maps and the displayed

monoid morphisms are implementations whose restrictions to nonempty words define the semantic bias (Lemma C.1).

LG is the pseudovariety of finite semigroups S in which every local monoid eSe, with e idempotent, is a group; classically $\mathbf{LG} = \mathbf{G} * \mathbf{D}$ [37, 38]. It is exactly the class of finite semigroups containing no copy of the two-element semilattice $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$) [36, Proposition 2.3]; that semilattice is the positive image of the typing behind the exponential lower bound of the companion CFG paper [25, Proposition 6.4]. Theorem 4.28 adds to this *known* algebra the sliding-factor group typing used for the MCFG quantitative bound.

Throughout, Σ is nonempty, $k \geq 1$, and A is a finite group. For a word w with $|w| \geq k+1$ let $\sigma_k(w)$ be the sequence of its factors of length $k+1$, ordered by their starting positions.

Definition 4.25 (Local-group typing)**.** Let $g : \Sigma^{k+1} \to A$ be any map. For $|w| \geq 2k$ put

$$\gamma_{k,g}(w) := g(F_1)g(F_2)\cdots g(F_{|w|-k}), \qquad \sigma_k(w) = (F_1, \ldots, F_{|w|-k}),$$

the ordered product in A, and put $\gamma_{k,g}(w) := 1_A$ for $|w| < 2k$. The *local-group typing* is $\lambda_{k,g}(w) := (\widehat{s}_{k,k}(w), \gamma_{k,g}(w))$. A finite typing $\widehat{s}$ is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,g}$ for some k, A and g.

For $\ell = k$, words of length $< 2k$ are *short* and recorded exactly, while a *long* w is recorded by $\langle \mathrm{pref}_k(w), \mathrm{suf}_k(w)\rangle$ (Definition 2.4). For $s, p \in \Sigma^k$ write $J_g(s,p) := \prod_{i=1}^{k} g((sp)[i..i+k])$, the ordered product over the k factors of length $k+1$ of sp that start in s.

Lemma 4.26 (Local-group typings are morphisms)**.** *The kernel of $\lambda_{k,g}$ is a congruence of finite index on Σ^*. Hence $\lambda_{k,g}$ is a homomorphism onto a finite monoid $M_{k,g}$. For long u, v,*

$$\lambda_{k,g}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,g}(u)\, J_g(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,g}(v)\big).$$

Proof. The window component is a morphism by Definition 2.4. For long u, v, the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the rest are those of v, so $\gamma_{k,g}(uv)$ is determined by u, $\mathrm{pref}_k(v)$ and $\gamma_{k,g}(v)$; the case of long u and short v is symmetric, and if both are short uv is determined exactly. Thus $\lambda_{k,g}(uv)$ is a function of $\lambda_{k,g}(u)$ and $\lambda_{k,g}(v)$, and the image has at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|A|$ elements. □

Lemma 4.27 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ be a factorization of a word into designated blocks β_j and fixed words x_j. Let $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$, where $\widehat{s}_{k,k}(\beta'_j) = \widehat{s}_{k,k}(\beta_j)$ for every j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$. If w is long, there are $c_0, \ldots, c_q \in A$, depending only on the x_j and the window types of the β_j, such that*

$$\gamma_{k,g}(w) = c_0\, \gamma_{k,g}(\beta_1)\, c_1 \cdots \gamma_{k,g}(\beta_q)\, c_q, \qquad \gamma_{k,g}(w') = c_0\, \gamma_{k,g}(\beta'_1)\, c_1 \cdots \gamma_{k,g}(\beta'_q)\, c_q.$$

Proof. A short block is recorded exactly, so $\beta'_j = \beta_j$, and we absorb it into the fixed words. Every remaining block is long and its replacement is long with the same first and last k letters; hence w' is long exactly when w is, with the same first and last k letters, which gives the window claim. A factor F of length $k+1$ of w either lies inside a long block β_j or meets each long block it touches only in a prefix or a suffix of length at most k (meeting both ends would force $|\beta_j| \leq k$). Factors of the second kind are spelled by the fixed words and the windows of the blocks, and occur in the same order in w'. Ordering all factors by starting position, those inside β_j form one consecutive run contributing $\gamma_{k,g}(\beta_j)$; let c_j be the ordered product of the remaining factors between consecutive runs. □

Theorem 4.28 (Characterizations of local-group positive images)**.** *For a finite typing $s : \Sigma^+ \to \mathsf{S}$ put $S := \widehat{s}(\Sigma^+)$. The following are equivalent.*

(i) *$S \in \mathbf{LG}$; that is, eSe is a group for every idempotent e of S.*

(ii) *S has no subsemigroup isomorphic to U_1.*

(iii) *Every idempotent of S lies in the minimal ideal of S.*

(iv) *$\widehat{s}$ is local-group-determined. One may take $k = |S|$.*

Proof. We include elementary arguments for the standard equivalence of (i)–(iii) (cf. [36, Proposition 2.3]), independently of the new sliding-factor statement (iv). Let K be the minimal ideal of S.

(i)⇒(ii). If $\{e, z\} \cong U_1$ with $ez = ze = z \neq e$, then z is a nonidentity idempotent in the group eSe, which is impossible.

(ii)⇒(iii). For an idempotent $e \in S$ choose $a \in K$ and put $z = (eae)^\omega$. Then z is an idempotent in $K \cap eSe$ with $ez = ze = z$. If $z \neq e$, the pair $\{e, z\}$ is a copy of U_1, a contradiction. Hence $e = z \in K$.

There is also a direct proof of (i)⇒(iii) that does not invoke the two-element semilattice: for any idempotent e and $a \in K$, the element eae lies in $K \cap eSe$; since eSe is a group with identity e, it has an inverse $y \in eSe$. Consequently $e = (eae)y \in K$, because K is an ideal.

(iii)⇒(i). The minimal ideal of a finite semigroup is completely simple [39]. For an idempotent $e \in K$, the ideal property gives $eSe = eKe$; this is the maximal subgroup (the $\mathcal{H}$-class) containing e, so it is a group.

(iii)⇒(iv). By the Rees theorem [39] write $K = \mathcal{M}(H; I, \Lambda; P)$, with product $(i, a, \lambda)(j, b, \mu) = (i, ap_{\lambda j}b, \mu)$, and put $n := |S|$. *Long products lie in K.* The Rees quotient S/K has no idempotent other than 0. In the Rees quotient S/K every element has a power equal to 0: its eventual idempotent power is necessarily the unique idempotent 0. For $s_1, \dots, s_m \in S$ with $m := |S/K| \leq n$, assume all m nonempty prefix products $p_1, \dots, p_m$ are nonzero in S/K. There are only $m - 1$ nonzero elements, so $p_i = p_j$ for some $i < j$. Writing $x = s_{i+1} \cdots s_j$ gives $p_i = p_i x$, hence $p_i = p_i x^\omega = 0$, a contradiction. Thus every product of at least n letters lies in K. *Right action.* Fix $i_0 \in I$ and for $\lambda \in \Lambda$ let $e_\lambda := (i_0, p_{\lambda i_0}^{-1}, \lambda)$, an idempotent with $xe_\lambda = x$ for every $x = (i, a, \lambda) \in K$. Writing $e_\lambda s = (j, c, \mu) \in K$ for $s \in S$ gives

$$(i, a, \lambda)\, s = (i,\ a\, \rho(\lambda, s),\ \mu(\lambda, s)), \qquad \rho(\lambda, s) := p_{\lambda j} c,$$

with $\rho(\lambda, s) \in H$ and $\mu(\lambda, s) \in \Lambda$ depending only on (λ, s). Right multiplication by s preserves the I-coordinate, as is explicit in this formula. More explicitly, since $(i, a, \lambda)e_\lambda = (i, a, \lambda)$ and $e_\lambda s = (j, c, \mu)$, associativity gives $(i, a, \lambda)s = (i, a, \lambda)(j, c, \mu) = (i, ap_{\lambda j}c, \mu)$, without requiring $s \in K$. Dually, for $z = (i, a, \lambda) \in K$ choose an idempotent $f_i \in K$ with $f_i z = z$. Then $sz = (sf_i)z$ and $sf_i \in K$, so the Rees product shows that left multiplication preserves the Λ-coordinate. In particular, for $|w| \geq n$ the Λ-coordinate of $\widehat{s}(w)$ is that of $\widehat{s}(\mathrm{suf}_n\, w)$, and the I-coordinate that of $\widehat{s}(\mathrm{pref}_n\, w)$. *Window factorization.* Let $w = a_1 \cdots a_m$ with $m \geq 2n$, and let λ_j be the Λ-coordinate of $\widehat{s}(a_{j-n+1} \cdots a_j)$ for $j \geq n$. Iterating the right action gives

$$\widehat{s}(w) = \Big(i,\ a \prod_{j=n}^{m-1} \rho(\lambda_j, \widehat{s}(a_{j+1})),\ \lambda_m\Big), \qquad \widehat{s}(a_1 \cdots a_n) = (i, a, \lambda_n),$$

and the j-th factor depends only on the factor $a_{j-n+1} \cdots a_{j+1}$ of length $n + 1$. With $g(F) := \rho(\lambda(\widehat{s}(F[1..n])), \widehat{s}(F[n+1]))$ for $F \in \Sigma^{n+1}$, the product is $\gamma_{n,g}(w)$. So $\widehat{s}(w)$ is determined by $\mathrm{pref}_n\, w$,

$\mathrm{suf}_n\, w$ and $\gamma_{n,g}(w)$; words of length $< 2n$ are recorded exactly by $\widehat{s}_{n,n}$, and ε is alone in its $\lambda_{n,g}$-class. Thus $\widehat{s} \preceq \lambda_{n,g}$ with $A = H$.

(iv)⇒(i). **LG** is closed under homomorphic images, so it suffices to show $T := \lambda_{k,g}(\Sigma^+) \in \mathbf{LG}$. A short element is never idempotent, since the square of a short word is longer. Let $e = (\langle p, s\rangle, \gamma_e)$ be idempotent. By Lemma 4.26, $\gamma_e J_g(s,p)\gamma_e = \gamma_e$, and for every $x \in T$, exe has window $\langle p, s\rangle$. Hence eTe lies in $H_e := \{(\langle p, s\rangle, \alpha) : \alpha \in A\}$, whose product $\alpha * \beta = \alpha J_g(s,p)\beta$ makes $\alpha \mapsto \alpha J_g(s,p)$ an isomorphism onto A. Thus eTe is a finite subsemigroup of a group, hence a group. □

Proposition 4.29 (Window–group typings are local-group typings, strictly). (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a homomorphism g into a finite group, then $\widehat{s} \preceq \lambda_{K,g'}$ with $K = \max\{k, \ell, 1\}$ and $g'(F) := g(\text{last letter of } F)$. Hence $\mathbf{LI} \vee \mathbf{G} \subseteq \mathbf{LG}$.*

(ii) *Let $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$. Let $h_{\mathtt{ab}} := \lambda_{1,g}$ with $g(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $g(F) = 0$ otherwise; it records the parity of the number of occurrences of $\mathtt{ab}$, together with the window. Then $h_{\mathtt{ab}}$ is not window–group-determined. Hence $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{G})$.*

Proof. (i) For $|w| \geq 2K$, $g(w) = g(\mathrm{pref}_K\, w)\, \gamma_{K,g'}(w)$, and shorter words are recorded exactly. For the inclusion, let $S \in \mathbf{LI} \vee \mathbf{G}$ and take $\Sigma = S$ with the evaluation typing $s : \Sigma^+ \twoheadrightarrow S$. Lemma 4.22 makes $\widehat{s}$ window–group determined, hence local-group-determined by the first part, so $S \in \mathbf{LG}$ by Theorem 4.28.

(ii) Suppose $h_{\mathtt{ab}} \preceq (\widehat{s}_{k,\ell}, g')$ with $g' : \Sigma^* \to A'$. Let $K = \max\{k, \ell, 1\}$ and let q be an even multiple of the exponent of A'. Put $u = \mathtt{a}^K(\mathtt{abc})^q\mathtt{a}^K$ and $v = \mathtt{a}^K(\mathtt{bac})^q\mathtt{a}^K$. Both are long, with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2K} = g'(v)$. But u contains q occurrences of $\mathtt{ab}$, while v contains exactly one, at $\mathtt{a}^K\mathtt{b}$, so $h_{\mathtt{ab}}(u) \neq h_{\mathtt{ab}}(v)$. Membership in **LG** follows from Theorem 4.28; if the positive image were in $\mathbf{LI} \vee \mathbf{G}$, Lemma 4.22 would make $h_{\mathtt{ab}}$ window–group-determined. □

Lemma 4.30 (Short tuples with prescribed local-group type). *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of fan-out at most f and rule rank at most $r_{\max} \geq 1$, and let $\lambda = \lambda_{k,g}$. Let $b_{\mathrm{w}} := f(2k+1)n_{\mathrm{w}}(r_{\max}+1)\bar{\theta}_G$, where $n_{\mathrm{w}} \leq |V||M_{k,k}|^f$ is the number of non-start typed nonterminals of the trimmed $\widehat{s}_{k,k}$-refinement. Put $n_\lambda := |V||M_{k,g}|^f$, and let $M_{A,f}$ be the constant of Section 4.4. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,g}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq \beta_\lambda := (r_{\max} + 1)\, b_{\mathrm{w}} \max\{1, (M_{A,f} - 1)\, n_\lambda\}.$$

If A is abelian, $M_{A,f} - 1$ may be replaced by $\mathtt{d}(A^f)$.

Proof. Let T be minimal as in Lemma 4.9 with root λ-vector $\mathbf{t}$ and threshold $b = b_{\mathrm{w}} \geq \bar{\theta}_G$. For a heavy node v labelled C, with yield $\vec{x}$ and window vector $\boldsymbol{\omega} = \widehat{s}_{k,k}(\vec{x})$, the typed nonterminal $C_{\boldsymbol{\omega}}$ is retained in the trimmed $\widehat{s}_{k,k}$-refinement, because T can be inserted into a successful derivation (G is reduced). Lemma 4.7 (trivial κ) gives a replacement tuple of C with the same window vector and length at most b_{w}. For pairwise incomparable heavy nodes $v_1, \ldots, v_m$ and $J \subseteq [m]$, let T_J replace the subtrees at v_i, $i \in J$. The components of the v_i are disjoint contiguous blocks of the root components. By Lemma 4.27, every T_J has the root window vector of T; a short root component is unchanged, and on each long root component $\gamma_{k,g}$ is an ordered product $c_0 y_1 c_1 \cdots y_q c_q$ of block values between constants that do not depend on J. The long root components thus form a block-replacement family over A (Definition 4.16). If $m \geq M_{A,f}$, or $m > \mathtt{d}(A^f)$ for abelian A, Lemma 4.17 gives a nonempty J with the root λ-vector of T, while T_J is strictly shorter. This contradicts minimality, and Lemma 4.9 with $N = M_{k,g}$ gives the bound. □

Detailed proof of Theorem 4.24. The theorem was stated above; we verify here the bound using the sliding-factor characterization and simultaneous ordered group-block replacement.

Proof of Theorem 4.24. By Theorem 4.28, $\widehat{s} \preceq \lambda_{k,g}$ for some k, A, g depending only on $\widehat{s}$. For a non-start typed nonterminal $X = B_{\mathbf{s}}$ and $\vec{u} \in L_X$, Lemma 4.30 gives $\vec{u}' \in L_B(G)$ with the same λ-vector and $\|\vec{u}'\|_1 \le \beta_\lambda$, and Lemma 4.3 gives $\vec{u}' \in L_X$. The role of the two shortening stages is distinct: the window stage provides replacements of length b_{w} without prescribing their group values, while the block-replacement lemma selects a nonempty family of such replacements whose *combined* ordered group products are unchanged. No per-subtree group-value preservation is assumed. Since $b_{\mathrm{w}} = O(\|G\|\bar{\theta}_G)$ and $n_\lambda = O(\|G\|)$, the heavy-node bound gives $\beta_\lambda = O(\|G\|^2\bar{\theta}_G)$ for fixed $f, r_{\max}, s, \Sigma$. There are $O(\|G\|)$ typed rules, and Lemma 4.2 then yields the displayed cubic bound on the *total* encoded positive-data length. □

By Proposition 4.29(i), this theorem also applies to window and window–group typings, but its general cubic bound is weaker than the quadratic bounds proved for those subclasses in Theorems 4.8 and 4.14.

Corollary 4.31 (Pseudovariety boundary; companion-based negative part)**.** *Let* $\mathbf{V}$ *be a pseudovariety of finite semigroups. Part* (ii) *uses the doubling-family theorem of the companion CFG paper [25, Proposition 6.4], available in arXiv version 5. Unlike part* (i)*, that lower-bound input is cited rather than reproved self-containedly here.*

(i) *If* $\mathbf{V} \subseteq \mathbf{LG}$*, then for every alphabet, every typing* $\widehat{s}$ *with* $\widehat{s}(\Sigma^+) \in \mathbf{V}$*, and all fixed* $f, r_{\max}$*, characteristic data of total length* $O(\|G\|^3\bar{\theta}_G)$ *exist as in Theorem 4.24.*

(ii) *If* $\mathbf{V} \not\subseteq \mathbf{LG}$*, then* $U_1 \in \mathbf{V}$*. At fan-out one there is a typing* h_{U} *with* $h_{\mathrm{U}}(\Sigma^+) = U_1$ *and targets with* $\|G\|, \theta_G = O(n)$*, every characteristic sample of which for* $\mathcal{B}^{\le 2}_{1,h_{\mathrm{U}}}$ *contains a word of length* $3 \cdot 2^n - 2$*. The same holds for every set-driven operator admitting characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,h_{\mathrm{U}}}$*.*

Proof. (i) is Theorem 4.24. For (ii), choose $S \in \mathbf{V} \setminus \mathbf{LG}$; by Theorem 4.28 it has a subsemigroup isomorphic to U_1, and pseudovarieties are closed under subsemigroups. Take the doubling family L_n, the typing h_{U} and the long type-1 word w^*_n of [25, Proposition 6.4], where L_n is shown to be Clark–Eyraud substitutable, hence $(1, h_{\mathrm{U}})$-tuple-substitutable. The type filter $h_{\mathrm{U}}^{-1}(\zeta)$ is $(1, h_{\mathrm{U}})$-tuple-substitutable by the remark preceding Proposition 2.9, so by Proposition 2.9(i) so is $L^-_n := L_n \cap h_{\mathrm{U}}^{-1}(\zeta) = L_n \setminus \{w^*_n\}$. If $K \subseteq L_n$ omits w^*_n, then $K \subseteq L^-_n$ and Proposition B.6 gives $L(\mathcal{B}^{\le 2}_{1,h_{\mathrm{U}}}(K)) \subseteq L^-_n$; thus every characteristic sample contains w^*_n. The set-driven statement is the distinguishing-pair argument of [25, Proposition 6.4(iv)] for the pair L_n, L^-_n. □

Remark 4.32 (Scope of the dichotomy). (i) Part (i) holds at every fixed fan-out; part (ii) is established only at fan-out one. The doubling family does not lift directly: L_2 is not $(2, h_{\mathrm{U}})$-tuple-substitutable, since the type-1 pairs $([, \mathtt{a}])$ and $([\mathtt{a},])$ share the context $[[\mathtt{aa}]\,\square_1\,\mathtt{a}\,\square_2\,]$ but have different distributions (checked exhaustively for $n \le 3$; under the finer typing recording each component's set of letters, still with semilattice image, the failure appears at $n = 3$). Whether part (ii) holds at fan-out $f \ge 2$ is the pseudovariety form of Open Problem 6.4.

(ii) The dichotomy concerns pseudovarieties. For an individual typing $\widehat{s}$ with $\widehat{s}(\Sigma^+) \notin \mathbf{LG}$, the class $\mathcal{C}^{\mathrm{mcf}}_{1,s}$ need not contain a coded copy of the doubling family, and whether every such typing forces superpolynomial characteristic data remains open.

4.6 Unrestricted Standard Input MCFGs

The preceding bounds use a reduced good presentation to make the quantitative witness argument transparent. Goodness, nondeletion, nonpermutation, nonmerging, and nonemptiness of tuple components need *not* be imposed on the grammar supplied for the target language. What matters when empty components are permitted is the cost of realizing each nonempty-component pattern.

Definition 4.33 (Component-pattern thickness)**.** For a productive nonterminal A of a standard MCFG H, let $\mathrm{supp}(\vec{u}) := \{i : u_i \neq \varepsilon\}$ and $\mathsf{Pat}_H(A) := \{\mathrm{supp}(\vec{u}) : \vec{u} \in L_A(H)\}$. Put

$$t_{\mathrm{pat}}(H) := \max_{\substack{A \text{ productive} \\ I \in \mathsf{Pat}_H(A)}} \min_{\substack{\vec{u} \in L_A(H) \\ \mathrm{supp}(\vec{u}) = I}} \|\vec{u}\|_1, \qquad \bar{t}_{\mathrm{pat}}(H) := \max\{1, t_{\mathrm{pat}}(H)\},$$

with the maximum over an empty set taken as zero. In particular, the all-empty pattern costs zero.

Proposition 4.34 (Good normalization controlled by pattern thickness)**.** *Fix $f, r \geq 1$. Let H be any standard fan-out-$\leq f$, rank-$\leq r$ MCFG with $L(H) \cap \Sigma^+ \neq \emptyset$, and let H_{red} be its trimmed presentation. There exists a reduced good MCFG G for $L(H) \cap \Sigma^+$ with the same fan-out and rank bounds and*

$$\|G\| = O_{f,r}(\|H\|), \qquad t_G \leq t_{\mathrm{pat}}(H_{\mathrm{red}}), \qquad \bar{\theta}_G = O_{f,r}(\|H\| \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})).$$

Proof. Use Proposition 3.9 and the stagewise tuple-language identities of Lemma D.1. Every retained nonterminal language is an image of a union of original support-pattern fibres under component deletion, permutation and concatenation. These maps do not increase total yield length, so a shortest representative of one contributing original pattern bounds each retained nonterminal's shortest yield by $t_{\mathrm{pat}}(H_{\mathrm{red}})$. Lemma 4.1(iii) then gives the thickness estimate. The precise fibre-transfer argument is in Appendix A. □

Corollary 4.35 (Total characteristic-data length from any standard MCFG)**.** *Fix $f, r \geq 1$, a finite alphabet Σ, and a fixed finite typing s. Let H be* any *finite standard fan-out-$\leq f$, rank-$\leq r$ MCFG (deleting rules, permutations, merging and empty components permitted) for $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. Put $n := \max\{1, \|H\|\}$ and $t := \bar{t}_{\mathrm{pat}}(H_{\mathrm{red}})$. The capped reconstruction operator admits a characteristic sample $K \subseteq L$ of cardinality $O_{f,r,s}(n)$ with*

$$\|K\|_+ = \begin{cases} O(n^3 t), & \text{if } s \preceq s_{k,\ell} \text{ or } \widehat{s} \preceq (\widehat{s}_{k,\ell}, g) \text{ for a fixed finite-group morphism } g, \\ O(n^4 t), & \text{if } \mathsf{S} \in \mathbf{LG}. \end{cases}$$

The hidden constants depend on the fixed alphabet, typing, rank, fan-out, and (where applicable) window and group, not on H; an optional datum ε costs one.

Proof. For a nonempty positive core apply Proposition 4.34 and then Theorem 4.8, 4.14, or 4.24, respectively. The size is $O(n)$ and the new thickness is $O(nt)$. If $L \subseteq \{\varepsilon\}$, use $K = L$. □

4.7 Computing Short Characteristic Data

The bounds above exhibit some short tuple and some short context. Call a sentence context of a typed nonterminal X *derivational* if it arises from a successful typed derivation by deleting one occurrence of an X-subtree; such contexts are admissible. All canonical contexts used in Sections 4.1–4.5 and Section 4.5.1 are derivational (Lemma 4.1(iv) and the proofs of Lemma 4.4 and Theorem 4.6).

Proposition 4.36 (Effective short characteristic data)**.** *Fix $f, r_{\max}$ and a finite typing $\widehat{s} : \Sigma^* \to \mathsf{S}^\sharp$. Given a reduced good MCFG G of fan-out at most f and rule rank at most $r_{\max}$, the following can be computed in time polynomial in $\|G\|$, with lengths written in binary:*

(i) $\theta_G(R)$ for every rule R, and hence θ_G and t_G;

(ii) the trimmed refinement $\widetilde{G}_0$ and, for each non-start typed nonterminal X, a canonical tuple $\omega(X) \in L_X$ of minimum total length and a derivational context $\chi(X)$ of minimum fixed length, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ belongs to $\widetilde{G}_0$;

(iii) derivation DAGs of size polynomial in $\|G\|$ representing these tuples and contexts, and hence every word of the witness set $\mathsf{W}(\widetilde{G}_0)$ of Definition 3.6 built from them.

Every word-length and total-length bound proved for $\mathsf{W}(\widetilde{G}_0)$ in Sections 4.1–4.5 and in Section 4.5.1 holds for this computed witness set. This does not apply to the relaxed samples $K^\dagger$ of Section 4.8, whose representatives need not be generated by their nonterminals. The words can be written out explicitly in time polynomial in $\|G\| + \|\mathsf{W}(\widetilde{G}_0)\|_+$.

Proof. For fixed $f, r_{\max}, s$ the full refinement $G^{\widehat{s}}$ has at most $|V||\mathsf{S}^\sharp|^f$ typed nonterminals and $|P||\mathsf{S}^\sharp|^{fr_{\max}}$ typed rules, and trimming is a linear-time productivity and reachability computation. View the typed rules as weighted deduction steps. A rule $X \to \rho(Y_1, \dots, Y_r)$ maps child weights to $(c_\rho, 1) + \sum_j (x_j, n_j)$, where c_ρ is the number of terminals written by ρ and pairs are compared lexicographically (terminal length, derivation-node count). This function is strictly superior to each argument even when no terminal is written, so the shortest-derivation algorithm in the weighted deductive form of Nederhof [35], building on Knuth's generalization of Dijkstra's method (see also Ramalingam and Reps [34]), computes $\lambda(X) = \min\{\|\vec{u}\|_1 : \vec{u} \in L_X\}$ together with an acyclic choice of minimizing rules, that is, a derivation DAG for $\omega(X)$. A second Dijkstra pass computes outside lengths, breaking equal-length ties by the number of edges: it starts with $\mu(X) = 0$ when $\widetilde{S} \to X$ is present, and for every typed rule $X \to \rho(Y_1, \dots, Y_r)$ and every j adds an edge from X to Y_j of weight $c_\rho + \sum_{i \neq j} \lambda(Y_i)$. Then $\mu(X)$ is the minimum fixed length of a derivational context of X, and the shortest-path tree supplies its DAG. The same two passes on G give $t_G = \max_A \lambda(A)$ and $\theta_G(R) = \mu(A) + c_\rho + \sum_j \lambda(B_j)$ for $R : A \to \rho(B_1, \dots, B_r)$.

Each of the cited proofs bounds a witness word through the length of its context, of its rule template and of its canonical child tuples, and does so by exhibiting *some* tuple of L_X and *some* derivational context. The computed minima are no longer, and the number of witnesses is unchanged, so every stated bound persists. Expanding a DAG takes time linear in the length of the word it represents. □

Proposition 4.37 (Rule-only witnesses and a computable data budget)**.** *In the setting of Proposition 4.36, choose all canonical tuples by the consistent acyclic minimum-weight derivations described there, and let $\mathcal{R}$ be the non-start typed rules of $\widetilde{G}_0$ (rank-zero rules included).*

(i) For each non-start typed nonterminal X, let $R_X : X \to \rho_X(Y_1, \dots, Y_q)$ be the first rule of its chosen minimizing derivation (including $q = 0$). Then $\omega(X) = \rho_X(\omega(Y_1), \dots, \omega(Y_q))$. Consequently

$$\mathsf{W}(\widetilde{G}_0) = \{\chi(\mathrm{lhs}\, R)[\rho_R(\omega(Y_1), \dots, \omega(Y_q))] : R \in \mathcal{R}\}, \qquad |\mathsf{W}(\widetilde{G}_0)| \le |\mathcal{R}| \le |P||\mathsf{S}^\sharp|^{fr_{\max}},$$

and this set remains characteristic for $\mathcal{B}_{f,s}^{\le r_{\max}}$ whenever the target is in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.

(ii) Let $\lambda(X)$ be the minimum total yield length from X and $\mu(X)$ the minimum fixed length of a derivational context of X (with $\mu(X) = 0$ for a direct start link). For $R : X \to \rho(Y_1, \dots, Y_q)$ put

$$b(R) := c_\rho + \sum_{j=1}^{q} \lambda(Y_j), \qquad \mathsf{B}_{\mathrm{rule}}(G, \widehat{s}) := \sum_{R \in \mathcal{R}} \big(1 + \mu(\mathrm{lhs}\, R) + b(R)\big).$$

Then $\|\mathsf{W}(\widetilde{G}_0)\|_+ \le \mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$, and the integer $\mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$ is computable in time polynomial in $\|G\|$ for fixed $f, r_{\max}, s$, its value being represented in binary. If $L(G)$ also contains ε and its nonempty core is treated separately, the ε datum adds at most one to the budget.

Proof. (i) An optimal derivation of X starts with R_X, and each of its child derivations is optimal, since replacing a nonoptimal child by the chosen minimum would decrease the lexicographic weight. The node-count tie-breaker makes the simultaneous choice acyclic, which gives the identity, also for rank zero. Hence $\chi(X)[\omega(X)]$ is *identical* to the rule word of R_X, and every other term of Definition 3.6 is a rule word; the two sets coincide as sets of terminal words, so completeness and the characteristic-sample property are unchanged. There is at most one word per typed rule. (ii) The rule word of $R : X \to \rho(Y_1, \dots, Y_q)$ has length exactly $\mu(X) + b(R)$. Summing $|w| + 1$ over the indexed family gives $\mathsf{B}_{\mathrm{rule}}(G, \widehat{s})$; passing to distinct words can only lower it. The values λ, μ and the number of typed rules are computed, with polynomial binary length, by Proposition 4.36. □

This rule-only budget is presentation-sensitive and can be much smaller than the worst-case polynomial bounds; it is an efficiently computable *upper certificate*, not a claim of a minimum-size characteristic sample or an improved worst-case exponent. For the singleton $\{\mathsf{c}^{2^n}\}$ after Theorem 4.6, explicit words are exponentially longer than $\|G\|$, but the derivation DAGs remain of polynomial size.

4.8 Distributional Representatives and Abelian-Filtered Targets

For an existential characteristic-sample bound one may instead choose short representatives of the same *typed distribution classes*, even when they have no derivation from the corresponding nonterminal.

Proposition 4.38 (Relaxed distributional witnesses)**.** *Let G be a reduced good fan-out-$\le f$, rank-$\le r$ presentation of $L = L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $\widetilde{G}_0$ be its trimmed yield-type refinement. For every non-start typed nonterminal X, let $[X]$ be the $\equiv_{L,s}$-class containing L_X, and put $\kappa_G(X) = \min_{\vec{v} \in [X]} \|\vec{v}\|_1$, $\kappa_G = \max_X \kappa_G(X)$. There exists a characteristic positive sample $K^\dagger$ for both the capped operator $\mathcal{B}^{\le r}_{f,s}$ (with cap at least the rank of G) and the uncapped $\mathcal{B}_{f,s}$ satisfying*

$$|K^\dagger| \le |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr},$$
$$\|K^\dagger\|_+ \le (|V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr})(1 + \theta_G + \|G\| + (r+1)\kappa_G).$$

The bound concerns the nonempty core; if $\varepsilon \in L$, add the sample word ε at encoded cost one.

Proof. For each $X = A_{\mathbf{t}}$ choose $\omega^\dagger(X) \in [X]$ of total length $\kappa_G(X)$, and let $\chi^\dagger(X) := E_A$ from Lemma 4.1(iv), of fixed length at most θ_G ($\Box_1$ for a start link). It accepts every tuple of L_X, hence also $\omega^\dagger(X) \equiv_{L,s} \vec{u} \in L_X$. Let $K^\dagger$ consist of the words $\chi^\dagger(X)[\omega^\dagger(X)]$, the rule words $\chi^\dagger(X)[\rho(\omega^\dagger(Y_1), \dots, \omega^\dagger(Y_q))]$ and the rank-zero rule words. By Lemma B.5 the rule tuple is equivalent to a tuple of L_X, so $K^\dagger \subseteq L$; it also has the $\widehat{s}$-type of X. Apply the generalized Proposition 3.8 with $\iota(X) = \omega^\dagger(X)$ and $\chi(X) = \chi^\dagger(X)$: the first family of sample words is its representative evidence and the rule-word family is precisely its rule evidence. Componentwise types agree by the typed

rules and Lemma B.5; all contexts are admissible and start-linked contexts are identities. The proposition supplies the capped and uncapped derivation simulations. Proposition B.6 gives the reverse inclusion for all larger positive samples. A nonterminal word has length at most $\theta_G + \kappa_G$ and a rule word at most $\theta_G + \|G\| + r\kappa_G$; sum the encoded costs. Unlike Proposition 4.37, the nonterminal words cannot in general be omitted. □

For fixed $f, r, \widehat{s}$, a family G_n with $\|G_n\| \max\{1, \theta_{G_n}\} = n^{O(1)}$ and superpolynomial characteristic-data requirements must therefore have $\kappa_{G_n} = n^{\omega(1)}$; this is a necessary condition, not a proof that such a family exists.

Theorem 4.39 (Polynomial data for abelian-filtered targets)**.** *Fix a finite nonempty alphabet Σ, a finite semigroup typing $s : \Sigma^+ \to \mathsf{S}$ and bounds $f, r \geq 1$. Write $\widehat{s}$ for the canonical completion and $\mathsf{S}^\sharp$ for its codomain. There are constants C, e depending only on Σ, s, f, r with the following property. Let A be any abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a morphism, $\mathsf{H} \leq \mathsf{A}$, $\gamma \in \mathsf{A}$, and $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$. If $L = \widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$ has a reduced good fan-out-$\leq f$, rank-$\leq r$ MCFG presentation G of $L \cap \Sigma^+$, then $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and admits a characteristic sample for $\mathcal{B}^{\leq r}_{f,s}$ with*

$$\|K^\dagger\|_+ \leq C\big(\|G\| \max\{1, \theta_G\}\big)^e.$$

The constants do not depend on $\mathsf{A}, \varphi, \mathsf{H}, \gamma$. The datum ε is included when $\varepsilon \in L$; the degenerate cases $L \subseteq \{\varepsilon\}$ are handled directly.

Proof. Membership in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ follows from Proposition 2.9 and the remark preceding it; the positive part has the assumed presentation, and ε is admitted by Start. By Proposition 4.38 it suffices to bound κ_G by a polynomial in $\|G\|\bar{\theta}_G$ with exponent depending only on $\Sigma, \widehat{s}, f, r$.

Let $\psi : \Sigma^* \to \mathbb{N}^\Sigma$ be the Parikh map, $\Psi(\vec{u}) = \sum_i \psi(u_i)$, and let $\overline{\varphi} : \mathbb{Z}^\Sigma \to \mathsf{A}$ be the group homomorphism through which φ factors; put $\Lambda = \{z \in \mathbb{Z}^\Sigma : \overline{\varphi}(z) \in \mathsf{H}\}$. Fix for each $A \in V$ a tuple $\omega_A \in L_A(G)$ with $\|\omega_A\|_1 \leq t_G \leq \theta_G$. The context E_A accepts every A-tuple, and all accepted words lie in one coset of H, so

$$\Psi(\vec{u}) - \Psi(\omega_A) \in \Lambda \qquad (\vec{u} \in L_A(G)). \tag{$*$}$$

For $R : A \to \rho(B_1, \ldots, B_q)$ let $c_R \in \mathbb{N}^\Sigma$ count its explicit letters and $\delta_R = \Psi(\omega_A) - c_R - \sum_j \Psi(\omega_{B_j})$. By (*) applied to $\rho(\omega_{B_1}, \ldots, \omega_{B_q}) \in L_A(G)$, $\delta_R \in \Lambda$, and $\|\delta_R\|_1 \leq T := (r+1)t_G + \|G\|$. Let $\Gamma_T = \{z \in \Lambda : \|z\|_1 \leq T\}$ and $\Lambda_T = \langle \Gamma_T \rangle$. Induction on derivations, using $\Psi(\vec{u}) - \Psi(\omega_A) = \sum_j (\Psi(\vec{u}_j) - \Psi(\omega_{B_j})) - \delta_R$ for $\vec{u} = \rho(\vec{u}_1, \ldots, \vec{u}_q)$, shows $\Psi(\vec{u}) - \Psi(\omega_A) \in \Lambda_T$ for every $\vec{u} \in L_A(G)$.

Now let $X = A_{\mathbf{t}}$ have arity $d \leq f$ and fix $\vec{u} \in L_X$. For each i choose, from a fixed semilinear representation of the Parikh image of the regular language $\widehat{s}^{-1}(t_i) \cap \Sigma^+$, a linear set $c_i + \sum_{p \in P_i} \mathbb{N}p$ containing $\psi(u_i)$; the number and size of all such generators depend only on Σ and $\widehat{s}$. The nonnegative integer system

$$\sum_{i=1}^{d} \Big(c_i + \sum_{p \in P_i} \nu_{i,p} p \Big) - \sum_{z \in \Gamma_T} (\mu^+_z - \mu^-_z) z = \Psi(\omega_A)$$

is feasible (use the Parikh vectors of the u_i and the Λ_T-membership of $\Psi(\vec{u}) - \Psi(\omega_A)$). It has $|\Sigma|$ equations, $O_{f,s,\Sigma}((T+1)^{|\Sigma|})$ variables, and coefficients and right-hand side of absolute value $O_{f,s,\Sigma}(T+1)$; in particular, each coordinate of $\Psi(\omega_A)$ is at most $t_G \leq \theta_G \leq T$ for the nonempty core. Papadimitriou's bound for small nonnegative integral solutions of equality systems [40] (a solution with all entries at most $n(ma)^{2m+1}$ for m equations, n variables and entries bounded by a) gives a solution whose entries are bounded by a polynomial in $T+1$ with exponent depending only on $|\Sigma|$. Realize each resulting Parikh vector by a nonempty word $v_i \in \widehat{s}^{-1}(t_i)$ and put $\vec{v} = (v_1, \ldots, v_d)$. Then $\|\vec{v}\|_1$ is polynomial in $T+1$, $\widehat{s}^{(d)}(\vec{v}) = \mathbf{t}$, and $\Psi(\vec{v}) - \Psi(\vec{u}) \in \Lambda_T \subseteq \Lambda$. For every sentence context

E, the words $E[\vec{v}]$ and $E[\vec{u}]$ have the same $\widehat{s}$-value and $\varphi(E[\vec{v}]) - \varphi(E[\vec{u}]) = \overline{\varphi}(\Psi(\vec{v}) - \Psi(\vec{u})) \in \mathsf{H}$; hence $\vec{v} \equiv_{L,s} \vec{u}$ and $\kappa_G(X) \leq \|\vec{v}\|_1$. Since $T \leq (r+1)\theta_G + \|G\|$, the claim follows, with parameters independent of the auxiliary abelian group. □

Theorem 4.39 applies even when the positive image of $\widehat{s}$ lies *outside* **LG**: it is a sufficient condition on the target language, not on the finite control, and it claims no polynomial-time construction for ineffectively presented auxiliary groups.

5 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability is the direct positive-data baseline [13, 14]. Under trivial typing the present framework reproduces that semantic class; finite typing refines which observed tuples may be identified while preserving compositional rule witnesses, and $\widehat{s}_{k,\ell}$ lifts the boundary information of Yoshinaka's CFG (k,ℓ) hierarchy [12] componentwise. Since every (f,s) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

5.1 Yoshinaka's Baseline and Prefix–Suffix Specialization

Definition 5.1 (Yoshinaka multidimensional substitutability)**.** Using the sentence-context sets $\mathcal{E}_d$ defined in Section 2, a language L is *fD-substitutable* if, for every $1 \leq d \leq f$, all $E, F \in \mathcal{E}_d$, and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], \quad E[\vec{y}], \quad F[\vec{x}] \in L \Longrightarrow F[\vec{y}] \in L$. Write $\mathsf{YSub}(f)$ for this semantic class and $\mathsf{YSL}(f,r) := \mathsf{YSub}(f) \cap \mathsf{MCFL}(f,r)$, where r bounds rule rank. We refer to the displayed four-instance implication as the *substitution implication*. Yoshinaka's original definition quantifies over every arity $d \leq f$, uses nonempty tuple components, and requires nonempty internal separators in d-contexts; footnote 3 separately discusses the stronger alternative that permits empty components and separators [14, Section 3.1 and footnote 3].

The trivial typing removes the componentwise type premise, so $\mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$. In Yoshinaka's word-level (k,ℓ) condition [12], with $u \in \Sigma^k$, $v \in \Sigma^\ell$ and $uy_1v, uy_2v \neq \varepsilon$,

$$w_1uy_1vw_1',\ w_1uy_2vw_1' \in L \quad \Longrightarrow \quad (w_2uy_1vw_2' \in L \iff w_2uy_2vw_2' \in L).$$

The compared fragments uy_iv are nonempty although the y_i may be empty, and equality of their $\widehat{s}_{k,\ell}$-values is exactly the boundary information used here, including exact identity of words shorter than $m_0 = \max\{1, k+\ell\}$. Hence $(1, \widehat{s}_{k,\ell})$-substitutability is precisely Yoshinaka's (k,ℓ) condition, and the analogous componentwise boundary requirement at higher fan-out is a definition of this paper, not a claim that Yoshinaka (2008) formulated an MCFG (k,ℓ) notion.

A language is *strictly 2-local* if membership is determined by permitted adjacent pairs in $\vdash w \dashv$, for fixed endmarkers.

Lemma 5.2 (Strictly 2-local constraints are boundary typed)**.** *If $L_{\mathrm{sl}} \subseteq \Sigma^*$ is strictly 2-local, then for every $f \geq 1$, $L_{\mathrm{sl}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$.*

Proof. Let equal-$\widehat{s}_{1,1}$-type nonempty tuples $\vec{x}, \vec{y}$ share an accepting context E_0. Internal adjacent pairs of each y_i are permitted because they occur in $E_0[\vec{y}]$. If $E[\vec{x}] \in L_{\mathrm{sl}}$, every adjacent pair of $E[\vec{y}]$ is either in fixed material, internal to a y_i, or crosses a component boundary. These are permitted respectively by $E[\vec{x}]$, by $E_0[\vec{y}]$, or by equality of the first/last letters encoded by the boundary type; end-marker pairs are unchanged. Thus $E[\vec{y}] \in L_{\mathrm{sl}}$, and symmetry finishes the proof. □

5.2 An Intrinsically Rank-Two Boundary-Typed Target

In this subsection, $s_{1,1}$ denotes the positive-word restriction $\widehat{s}_{1,1}|_{\Sigma_{\mathrm{intr}}^+}$ of the prefix–suffix typing from Definition 2.4, with alphabet Σ_{intr} defined below.

Let

$$\Sigma_{\mathrm{intr}} := \{\mathtt{a},\mathtt{b},\mathtt{l},\mathtt{r},\mathtt{p},\mathtt{q},\mathtt{c},\mathtt{d},\#,\$,\mathtt{u},\mathtt{v},\mathtt{w}\}, \qquad P_{\mathrm{intr}} := \{\mathtt{lpc},\mathtt{lpd},\mathtt{lqc},\mathtt{rpc},\mathtt{rpd}\},$$

$$\mathrm{MIX}_2^+ := \{x \in \{\mathtt{a},\mathtt{b}\}^+ : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}, \qquad T_3^+ := \{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}, \qquad L_{\mathrm{intr}} := P_{\mathrm{intr}} \# \mathrm{MIX}_2^+ \,\$\, T_3^+.$$

The MIX_2^+ factor carries the rule-rank obstruction, which is not created by finite typing: $\mathrm{MIX}_2^+ = \{\mathtt{a},\mathtt{b}\}^+ \cap \varphi_{\mathrm{mix}}^{-1}(0)$ for $\varphi_{\mathrm{mix}}(\mathtt{a}) = 1$, $\varphi_{\mathrm{mix}}(\mathtt{b}) = -1$ in $\mathbb{Z}$, so Proposition 2.9(ii) already places it in $\mathsf{YSub}(f)$ for every f. The T_3^+ factor forces minimum fan-out two, and the wrapper P_{intr} separates the target from $\mathsf{YSub}(f)$.

Theorem 5.3 (A non-context-free intrinsic-rank-two typed witness)**.** *The language L_{intr} satisfies $L_{\mathrm{intr}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$, and, for every $f \geq 2$, $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}} \setminus \mathsf{YSub}(f)$. Moreover, for every finite $f' \geq 1$, $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f',1)$. Consequently the minimum fan-out of L_{intr} is exactly two, and its minimum rule rank among all finite-fan-out MCFG presentations is exactly two. In particular, Theorem 4.8 applies to a non-context-free language that cannot be covered by Theorem 4.6 at any finite fan-out.*

Proof. Grammar. Let B have the rules

$$B \to \mathtt{a}B\mathtt{b}B \mid \mathtt{b}B\mathtt{a}B \mid \mathtt{a}B\mathtt{b} \mid \mathtt{ab}B \mid \mathtt{ab} \mid \mathtt{b}B\mathtt{a} \mid \mathtt{ba}B \mid \mathtt{ba};$$

the first-return induction on $|x|_{\mathtt{a}} - |x|_{\mathtt{b}}$ gives $L_B = \mathrm{MIX}_2^+$. Let the fan-out-two nonterminal A have rules $A \to (\mathtt{u},\mathtt{w}) \mid (\mathtt{u}\xi_1^1\mathtt{v},\mathtt{w}\xi_1^2)(A)$, so $L_A = \{(\mathtt{u}^n\mathtt{v}^{n-1},\mathtt{w}^n) : n \geq 1\}$, and for every $\alpha \in P_{\mathrm{intr}}$ add $S \to \alpha\#\xi_1^1\$\xi_2^1\mathtt{v}\xi_2^2(B,A)$. The grammar generates L_{intr}, is reduced, λ-free, linear, nondeleting and nonpermuting, and is nonmerging because the two components of A are separated by the literal $\mathtt{v}$. It is good, of fan-out two and rule rank two.

Non-context-freeness. With $R := \mathtt{lpc}\#\mathtt{ab}\$\,\mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$, $L_{\mathrm{intr}} \cap R = \{\mathtt{lpc}\#\mathtt{ab}\$\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$, which a homomorphism erasing the fixed prefix maps onto the non-context-free $\{\mathtt{u}^n\mathtt{v}^n\mathtt{w}^n : n \geq 1\}$. By closure of CFL under intersection with regular languages and homomorphism, L_{intr} is not context-free; fan-out-one MCFGs are exactly CFGs here, so the minimum fan-out is two.

Typed membership. Let $Q_{\mathrm{intr}} := P_{\mathrm{intr}} \# \{\mathtt{a},\mathtt{b}\}^+ \,\$\, \mathtt{u}^+\mathtt{v}^+\mathtt{w}^+$. It is strictly 2-local, with the allowed adjacent pairs

$$\vdash \mathtt{l},\ \vdash \mathtt{r},\ \mathtt{lp},\ \mathtt{lq},\ \mathtt{rp},\ \mathtt{pc},\ \mathtt{pd},\ \mathtt{qc},\ \mathtt{c}\#,\ \mathtt{d}\#,\ \#\mathtt{a},\ \#\mathtt{b},\ \mathtt{aa},\ \mathtt{ab},\ \mathtt{ba},\ \mathtt{bb},$$
$$\mathtt{a}\$,\ \mathtt{b}\$,\ \$\mathtt{u},\ \mathtt{uu},\ \mathtt{uv},\ \mathtt{vv},\ \mathtt{vw},\ \mathtt{ww},\ \mathtt{w} \dashv .$$

Let $\Phi_{\mathrm{intr}} : \Sigma_{\mathrm{intr}}^* \to \mathbb{Z}^3$ send $\mathtt{a},\mathtt{b},\mathtt{u},\mathtt{v},\mathtt{w}$ to $(1,0,0)$, $(-1,0,0)$, $(0,1,0)$, $(0,-1,1)$, $(0,0,-1)$ and all other letters to 0. Its three coordinates impose $|x|_{\mathtt{a}} = |x|_{\mathtt{b}}$, $|x|_{\mathtt{u}} = |x|_{\mathtt{v}}$ and $|x|_{\mathtt{v}} = |x|_{\mathtt{w}}$, so $L_{\mathrm{intr}} = Q_{\mathrm{intr}} \cap \Phi_{\mathrm{intr}}^{-1}(0)$. Lemma 5.2 and Proposition 2.9(ii) make this intersection $(f, s_{1,1})$-tuple-substitutable for every f; with the grammar above, $L_{\mathrm{intr}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$ for every $f \geq 2$.

Separation. For $x = \mathtt{p}$, $y = \mathtt{q}$, $E = \mathtt{l}\square_1\mathtt{c}\#\mathtt{ab}\$\mathtt{uvw}$ and $F = \mathtt{r}\square_1\mathtt{d}\#\mathtt{ab}\$\mathtt{uvw}$, we have $E[x], E[y], F[x] \in L_{\mathrm{intr}}$ but $F[y] \notin L_{\mathrm{intr}}$, so the arity-one substitution implication fails and $L_{\mathrm{intr}} \notin \mathsf{YSub}(f)$ for every $f \geq 1$; the boundary typing separates the offending factors, $\widehat{s}_{1,1}(\mathtt{p}) \neq \widehat{s}_{1,1}(\mathtt{q})$.

Intrinsic rule rank. Gallot calls an MCFG *non-branching* when every positive-rank rule has a single child nonterminal, with no global fan-out bound [21, Definition 38]; his class is therefore

$\bigcup_{f'<\infty} \mathsf{MCFL}(f', 1)$. Definition 38 permits a child variable to occur at most once, and Theorem 28 shows that non-permuting non-erasing presentations generate the same class. Theorem 29 identifies the class with finite-index EDT0L, and Theorem 31 (via Propositions 5–6) shows that $\mathrm{MIX}_2 := \{x \in \{\mathtt{a}, \mathtt{b}\}^* : |x|_{\mathtt{a}} = |x|_{\mathtt{b}}\}$ is not finite-index EDT0L [21, Theorems 28, 29, 31 and Propositions 5–6]; Independently, the published Theorem C of Bishop et al. [22] excludes an EDT0L word problem for $\mathbb{Z}$ and thereby directly excludes MIX_2 (which is that word problem over generators $\mathtt{a}, \mathtt{b}$). Gallot's Definition 38 and Theorem 29 are still needed for the identification of non-branching MCFG languages with finite-index EDT0L. Let π fix $\mathtt{a}, \mathtt{b}$ and erase all other letters, so $\pi(L_{\mathrm{intr}}) = \mathrm{MIX}_2^+$. Applying π to the terminal material of a fan-out-f' rank-one presentation of L_{intr} would give one of MIX_2^+ (empty terminal pieces are allowed in the standard formalism), and a fresh start symbol with a unary identity link and a rank-zero ε rule would give one of MIX_2, a presentation in Gallot's class (Definition 38, Theorem 28), contradicting his lower bound. Hence $L_{\mathrm{intr}} \notin \mathsf{MCFL}(f', 1)$ for every finite f', and the displayed grammar proves the claimed intrinsic rule rank. □

The two lower bounds arise from separate factors rather than from an irreducible interaction between the resources, and the non-branching lower bound itself is prior work. What is specific here is that a strictly local wrapper and the $(1, 1)$ boundary typing place the target inside the typed class while the substitution implication keeps it outside Yoshinaka's trivial-typing baseline.

The quantitative gap between one- and higher-arity substitutability has a distinct source: in a two-hole context, a fixed separator can slide across different occurrences in one sentence. Repeated type pairs then force *distributional* equality, a constraint invisible at arity one. The next section makes this obstruction precise for the companion's doubling family without claiming a higher-arity characteristic-data lower bound.

6 A Higher-Arity Obstruction to Transferring the CFG Lower Bound

For a word m and positions $0 \le p < q \le |m|$, we write $m[0 : p]$ and $m[q : |m|]$ for its prefix and suffix. Only nonempty tuple components are used in the following lemma.

Lemma 6.1 (Sliding separators)**.** *Let $s : \Sigma^+ \to \mathsf{S}$ be a finite-semigroup typing and let L be $(2, s)$-tuple-substitutable. Suppose $v_0 m v_2 \in L$ and a nonempty word z occurs in m at $N > |\mathsf{S}|^2$ distinct positions p, each satisfying $0 < p$ and $p + |z| < |m|$. Then there are positions $p < q$ for which $\vec{x} = (m[0 : p], m[p + |z| : |m|]), \qquad \vec{y} = (m[0 : q], m[q + |z| : |m|])$ have equal componentwise s-types and $\mathcal{D}_L(\vec{x}) = \mathcal{D}_L(\vec{y})$.*

Proof. The two components of each tuple are nonempty, and all the tuples share the admissible context $E = v_0 \Box_1 z \Box_2 v_2$ with $E[\vec{x}] = E[\vec{y}] = v_0 m v_2 \in L$. There are at most $|\mathsf{S}|^2$ componentwise type pairs, so two distinct positions give equal types. By $(2, s)$-tuple substitutability and the common accepting context, their full sentence-context distributions agree. □

We apply the sliding effect to the bracketed doubling family of the revised companion CFG work [25]. The construction itself is repeated here, and the nontransfer proposition below is proved independently of the companion lower-bound theorem. To make the scope of the claim explicit, recall its construction. Let $\Gamma_{\mathrm{br}} = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$ and put $E_0 \to \mathtt{a}$ and

$$E_i \to \lfloor E_{i-1} E_{i-1} \rfloor \mid \lceil E_{i-1} \rceil \quad (1 \le i \le n), \qquad S_0 \to E_n.$$

Let L_n be the generated finite language, $w_0^* = \mathtt{a}$, $w_i^* = \lfloor w_{i-1}^* w_{i-1}^* \rfloor$, and $L_n^- = L_n \setminus \{w_n^*\}$. The following nonmembership result does *not* depend on the two-element semilattice typing used in the CFG lower bound.

Proposition 6.2 (No fixed finite typing lifts the doubling family)**.** *For every finite-semigroup typing $s : \Gamma_{\mathrm{br}}^+ \to \mathsf{S}$ and every $n \geq 2|\mathsf{S}| + 3$, neither L_n nor L_n^- is $(2, s)$-tuple-substitutable. Consequently, for every $f \geq 2$, $L_n, L_n^- \notin \mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

Proof. Put $q = |\mathsf{S}|$, and let $a := s(\lfloor)$. Among $a, a^2, \ldots, a^{q+1}$ choose equal powers $a^i = a^j$ with $1 \leq i < j \leq q + 1$. Writing $\pi = j - i$, the identity $a^{i+\pi} = a^i$ implies $a^{t+\pi} = a^t$ for every $t \geq i$. Our lower bound on n ensures $n - j - 1 \geq q + 1 \geq i$ and $j \leq n - 2$. It follows that $s(\lfloor^i) = s(\lfloor^j)$, $\quad s(\lfloor^{n-i-1}) = s(\lfloor^{n-j-1})$. Thus the nonempty pairs $\vec{x} = (\lfloor^i, \lfloor^{n-i-1})$, $\quad \vec{y} = (\lfloor^j, \lfloor^{n-j-1})$ have equal componentwise types.

For L_n take $w = w_n^*$. For L_n^- take w obtained from w_n^* by replacing a deepest binary rule on the rightmost branch by its unary $\lceil E_0 \rceil$ alternative. This replacement lies outside the leftmost branch, so in both cases $w = \lfloor^n v$ for a suffix v. The admissible context $E = \Box_1 \lfloor \Box_2 v$ fills with both $\vec{x}$ and $\vec{y}$ to produce the same word w in the respective target.

Now use the binary node on the leftmost branch at depth i, whose two children are instances of E_{n-i-1}. Replace that rule in w by its unary $\lceil E_{n-i-1} \rceil$ alternative, retaining the original left child and dropping the right child. Call the resulting word w'. The replacement is grammatical, and $w' \in L_n^- \subseteq L_n$. Since w' begins with $\lfloor^i \lceil \lfloor^{n-i-1}$, it has a factorization $w' = \lfloor^i \lceil \lfloor^{n-i-1} v'$ for some v'. The context $F = \Box_1 \lceil \Box_2 v'$ therefore accepts $\vec{x}$ in both target languages.

By contrast, $F[\vec{y}] = \lfloor^j \lceil \lfloor^{n-j-1} v'$. Relative to w', the $\lceil$ opening symbol has moved $j - i > 0$ positions into the previously contiguous initial sequence of $\lfloor$ openings, while the corresponding closing symbols in v' have not moved. The first $\rceil$ that matches this unary opening in w' is now preceded by too many $\rfloor$ closings for the $n - j - 1$ $\lfloor$ openings following $\lceil$; a $\rfloor$ must close $\lceil$. Hence the bracket types do not match, so $F[\vec{y}] \notin L_n$ and, a fortiori, $F[\vec{y}] \notin L_n^-$. We have an accepted common context, equal types, and different distributions, contrary to $(2, s)$-tuple substitutability. □

Remark 6.3 (What the obstruction does not prove). Proposition 6.2 strengthens the non-transfer observation of Remark 4.32: no *fixed* finite typing repairs two-dimensional substitutability for this particular doubling family when n grows. It is an instance of the sliding-separator phenomenon in Lemma 6.1. This rules out a direct reuse of the companion's examples; it does not rule out other higher-fan-out families with superpolynomial characteristic data. The family is not itself a higher-fan-out lower bound.

Open Problem 6.4 (Original-thickness bounds beyond fan-out one). Fix $f \geq 2$ and a rule-rank bound $r \geq 2$. Does there exist a fixed finite-semigroup typing s and a family of reduced good fan-out-$\leq f$, rank-$\leq r$ grammars G_n of targets in $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ with $\|G_n\| \max\{1, \theta_{G_n}\} = n^{O(1)}$ but superpolynomial total positive length for every characteristic sample of $\mathcal{B}_{f,s}^{\leq r}$? The family above cannot provide such examples, and Proposition 4.38 shows that any such family must have superpolynomial minimum typed-distributional representative length κ_{G_n}.

7 Conclusion

Qualitative positive-data identification of typed MCFGs follows Yoshinaka's distributional framework. We obtain quantitative bounds on the total encoded length of characteristic samples relative to the original size and rule thickness of reduced good MCFG presentations. Prefix–suffix window typing, and window typing combined with an arbitrary finite group, admit $O(\|G\|^2 \bar{\theta}_G)$ bounds.

The group proof simultaneously preserves ordered products across output components without assuming commutativity. For positive-image local-group typings we establish $O(\|G\|^3\bar{\theta}_G)$. Even at fan-out one, this strictly extends the $\mathbf{LI} \vee \mathbf{Grp}$ sufficient regime from the companion CFG work. We additionally establish bounds for rank-one grammars and abelian-filtered targets and give a computable rule-only data budget.

These algebraic hypotheses are sufficient conditions, and the displayed degrees have not been shown optimal. A fan-out-one lower-bound construction from the public companion CFG paper [25, Proposition 6.4] gives a companion-based pseudovariety boundary consequence. The construction's exponential family itself is explicit, and cannot be transferred to two-dimensional substitutability under any fixed finite typing. Distinguishing-pair arguments give an operator-independent forcing-depth criterion. Short Parikh-difference certificates show that abelian-coset competitors alone cannot yield a superpolynomial hitting-set obstruction under polynomial original rule thickness. For finite targets we give a tuple-nesting obstruction, a linear short-fibre bound in an interior-marker case, and a general finite-MCFG length-spectrum bound. These partial results do not settle whether higher fan-out can require superpolynomial characteristic-data length or forcing depth.

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Data availability

No empirical datasets were used or generated. All statements of the main mathematical results are supported by their proofs in the manuscript or by the cited literature. Finite-case computational checks serve only as supplementary consistency tests and are not prerequisites for the general theorems.

Methodological note on AI-assisted proof exploration

During the research process, ChatGPT (OpenAI) and Claude (Anthropic) were used to explore candidate arguments, check proof drafts for possible gaps, search for potential counterexamples, and compare citations and finite illustrative cases. Candidate arguments were incorporated only where they could be expressed as mathematical proofs in the main text or appendices. Computer-assisted checks of finite configurations are sanity checks rather than substitutes for general proofs. This research use is distinct from the manuscript-preparation disclosure immediately before the references.

A Pattern-Thickness Transfer for General MCFG Presentations

Here are the support-fibre details behind Proposition 4.34. Write $F_{A,I} := \{\vec{u} \in L_A(H_{\mathrm{red}}) : \mathrm{supp}(\vec{u}) = I\}$. Every realized fibre has a shortest tuple, of length at most $t_{\mathrm{pat}}(H_{\mathrm{red}})$.

For clarity, the source-language invariant is preserved *through all four stages*: each intermediate decorated nonterminal is the image of a union of realizable original fibres $F_{A,I}$ under one composition of projection, permutation and consecutive-component merging; where an intermediate support is prescribed, retain only the original fibres whose image has that support. The map is length-nonincreasing, and every such fibre has an actual derivation in the trimmed input. The stagewise bookkeeping is:

Stage	Tuple-language operation	Support/length information
(a) Nondeletion	$\pi_J(L_A)$, union over source fibres	Omitted children have derivations; projection does not increase length
(b) Nonpermutation	$\sigma(\pi_J(L_A))$	Support is permuted; length is unchanged
(c) Empty-pattern removal	Restrict to a realized support I, then π_I	Select preimage source fibres with the required support; length does not increase
(d) Nonmerging	$\mu(\sigma\pi_J(L_A))$, with the support restriction inherited from (c)	Merged support is a function of previous support; length is unchanged

In this table, L_A refers to the original input nonterminal and the operations are understood as applying to the appropriate union of $F_{A,I}$, not to an unrestricted L_A after stage (c). The original rank and fan-out bounds survive because the four constructions introduce only finitely many component decorations per rule for fixed f, r.

At stage (a) of Lemma D.1, a new nonterminal has language $\pi_J(L_A(H_{\mathrm{red}}))$, a union of the images $\pi_J(F_{A,I})$. Stage (b) permutes components and thus permutes support patterns without changing total length. At stage (c), restricting to a realized output pattern selects the images of those original support fibres whose projected/permuted support is exactly that pattern. Because each such original fibre is nonempty, selecting its shortest tuple yields an output tuple of total length at most $t_{\mathrm{pat}}(H_{\mathrm{red}})$. Stage (d) merges consecutive components; concatenation cannot increase total length, and its output support is determined by the input support, so the same argument applies after merging. The stagewise equalities are equalities of tuple languages, not merely inclusions; hence each retained nonterminal has an actual bounded-length terminal derivation. Fresh nonterminal names are only the finite decorators of these source nonterminals. Final trimming removes symbols without changing the tuple languages of the surviving ones. This proves $t_G \le t_{\mathrm{pat}}(H_{\mathrm{red}})$.

The normal form has $\|G\| = O_{f,r}(\|H\|)$ by Proposition 3.9; consequently $\bar{\theta}_G \le 2\max\{1, \|G\|\}\max\{1, t_G\} = O_{f,r}(\|H\|\bar{t}_{\mathrm{pat}}(H_{\mathrm{red}}))$ by Lemma 4.1(iii). This argument does not assert polynomial control by the *ordinary* untyped rule thickness of H: Proposition D.3 shows why elimination of empty components can be expensive.

B Auxiliary Verification of Qualitative Reconstruction

B.1 Qualitative Learning via Yoshinaka (2015)

Qualitative learning for fixed rule rank is inherited from Yoshinaka's signature theorem [15, Theorem 1]. We record the specialization to our typed grammar and its actual implementation, rather

than repeat the general learning proof. Fix $f, r \geq 1$ and $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$. Let $\mathfrak{S}_{f,r,s}$ have the initial sort $O_* = \{\overline{w} : w \in \Sigma^*\}$ and tagged, pairwise disjoint sorts

$$T_{d,\boldsymbol{\mu}} = \{\vec{w} \in (\Sigma^+)^d : s^{(d)}(\vec{w}) = \boldsymbol{\mu}\}, \quad 1 \leq d \leq f, \quad \boldsymbol{\mu} \in \mathsf{S}^d.$$

Its functions are all type-compatible good MCFG templates of rank at most r (including constants), the promotions $\pi_\mu : (w) \mapsto \overline{w}$ to O_*, and the initial constant $\overline{\varepsilon}$. No operation takes an O_* input. This is the nonempty-component variant of the MCFG signature of [15, Example 3 and Section 8] with a finite refinement of sorts.

Proposition B.1 (Fixed-rank specialization)**.** *For each noninitial sort $T_{d,\boldsymbol{\mu}}$, the realizable substructures are exactly $T_{d,\boldsymbol{\mu}}$ and the realizable contexts are the functions $\vec{x} \mapsto \overline{E[\vec{x}]}$ for $E \in \mathcal{E}_d$; the only context of O_* is the identity. Moreover,*

$$\mathsf{Subst}(\mathfrak{S}_{f,r,s}) = \mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r).$$

For fixed f, r, s, the signature meets Assumptions 1–3 of [15].

For the fixed set of tagged sorts, the template functions preserve sorts by the fixed multiplication table of S. Finite observed substructures and their contexts are enumerable by finite cut positions. The remaining signature axioms follow as in [15, Example 3 and Section 8]; this comparison is used for qualitative attribution only.

Remark B.2 (Relation to the general signature framework). The witnessed Link and Comp rules implement a substructure-indexed reconstruction in the spirit of Yoshinaka's $\mathrm{SUBST_P}$ [15, Section 6.1]. We use this comparison for attribution, not as a formal identification of the two grammar codes or as a premise for soundness or completeness. Those properties are proved directly for $\mathcal{B}_{f,s}$.

Corollary B.3 (Inherited qualitative learning at fixed rank)**.** *For fixed f, r, s, the class $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r)$ is identifiable in the limit from positive data with polynomial-time hypothesis updates. For a reduced target signature grammar with N nonterminals and P rules, Yoshinaka's $\mathrm{SUBST_D}$ has a characteristic set of at most $|N| + |P|$ words.*

Proof. Apply [15, Theorem 1 and the paragraph following Lemma 3] to Proposition B.1. □

Remark B.4 (Boundary between inherited and new results). The explicit proof of [15, Theorem 1] concerns $\mathrm{SUBST_D}$. Its Section 7 also discusses $\mathrm{SUBST_P}$, but we do not import a claimed equality of that operator with the concrete grammar $\mathcal{B}_{f,s}$. In particular, the direct soundness and characteristic-witness arguments below do not use such an equality. Characteristic-set *cardinality* does not bound its *total encoded length*, the object of Section 4. A single fixed-rank signature also does not cover unbounded rule rank; the concrete finite-witness argument below extends [14, Corollary 2] to typed links.

B.2 Soundness

Lemma B.5 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,s} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \ldots, \vec{u}_r) \equiv_{L,s} \rho(\vec{x}_1, \ldots, \vec{x}_r)$.*

Proof. Replace the children one at a time, keeping all other children fixed and nonempty. Given an accepting outer context E for the result of any intermediate replacement, Lemma 3.1 induces an admissible context E_j of the child being replaced. Since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, membership in L is unchanged under that replacement, in either direction. The same argument applies at each step even if earlier

children have already been replaced. Finally, the componentwise s-types of the output tuples agree because the template is fixed and $\widehat{s}$ is multiplicative on concatenation, including the fresh identity for empty terminal gaps. Therefore the output distributions coincide. □

Proposition B.6 (Soundness of every hypothesis)**.** *Let L be (f,s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \geq 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\leq r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proof. Proceed by induction on the height of a derivation rooted at $[\vec{x}]$. A Const rule derives exactly $\vec{x}$. For Link, $[\vec{x}] \to [\vec{y}]$ is witnessed by a common admissible context of same-type tuples in $K \subseteq L$, hence tuple substitutability gives $\vec{x} \equiv_{L,s} \vec{y}$; compose this equivalence with the inductive hypothesis for $[\vec{y}]$. For Comp, each child derivation produces a tuple equivalent to its witnessed child, and Lemma B.5 transfers those equivalences through the witnessed template. Its original output is exactly the observed parent $\vec{x}$, establishing the invariant. Every positive Start rule points to an observed sentence (w), whose derivational context is the identity; the invariant implies $w \in L$. The empty Start rule is included only for $\varepsilon \in K \subseteq L$. This argument holds at every finite rank cap; for nonempty K the uncapped construction has cap $\|K\|_+$, and the empty case is immediate. □

B.3 Effective Reconstruction and Conservative Learning

Theorem B.7 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and s, and implement type tests by the completion $\widehat{s}$. The encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Proof. Put $n_K = \max\{1, \|K\|_+\}$. An occurrence is a sample word with at most $2f$ cut positions, so there are $n_K^{O(f)}$ occurrences, typed by fixed-monoid products; grouping occurrences by context and type lists the Link rules. A composition witness of rank $r \leq r_{\max}$ is determined by a parent occurrence and at most $2f r_{\max}$ further cut positions inside it, with a labelling whose number depends only on $f, r_{\max}$; goodness is checked directly. This gives $n_K^{O(f r_{\max})}$ candidate Comp rules; the remaining rules have rank at most one, and deduplication leaves an encoding of the same size. □

Definition B.8 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}_{f,s}^{\leq r_{\max}}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}_{f,s}^{\leq r_{\max}}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary B.9 (Gold identification)**.** *Fix $f \geq 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}_{f,s}^{\leq r_{\max}}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$, in time per datum polynomial in the size of the content read so far.*

Proof. For fixed rule rank, qualitative identification follows from the direct soundness and exact-reconstruction results (Proposition B.6 and Theorem 3.10) with the conservative policy of [14, Algorithm 1]; the finite-signature attribution is Corollary B.3. For unbounded rank, Theorem 3.10 provides a finite characteristic sample, while Proposition B.6 gives soundness. Membership is decidable [29]. After a text has presented the characteristic sample, the next rebuild (if any) is correct and permanent; if no rebuild follows, sample consistency and conservativeness imply that the current hypothesis already generates every text word and hence L. For fixed rank, the per-datum bound follows from Theorem B.7 and [14, Proposition 4]. □

Update complexity versus convergence stage. The polynomial-time guarantee of Theorem B.7 and Corollary B.9 concerns hypothesis reconstruction and membership testing at each finite stage, measured against the data seen so far. TxtEx convergence in Corollary B.9 does not bound the first stage of stabilization or the number of mind changes polynomially: a text may postpone the first presentation of a particular positive example arbitrarily long. Short characteristic data by itself gives no presentation-independent convergence-stage bound.

Every CFG has a rank-two presentation, so $\mathcal{A}_{1,s}^{\leq 2}$ covers $\mathcal{C}_{1,s}^{\mathrm{mcf}}$; this does not imply polynomial characteristic-data length at rank two (Section 4).

Proof of Result B.1. Rank-zero constants realize every noninitial substructure. By Lemma 3.1, contexts induced through good templates remain admissible. Conversely, every $E = v_0\Box_1 v_1 \cdots \Box_d v_d \in \mathcal{E}_d$ is realized by the good unary template $(v_0\xi_1^1 v_1 \cdots \xi_1^d v_d)$ followed by promotion. This identifies the signature condition with Definition 2.7. Erasing tags yields rank-$\leq r$ MCFGs; the converse uses rank-preserving good normalization and yield-type refinement (Propositions 3.9 and 3.5), with $\bar{\varepsilon}$ handling the empty word. The operation arities are bounded by r. For fixed f, r, s, types, contexts and grammar membership are polynomial-time decidable [14, Proposition 4]; substructures, contexts and witnessed typed function names (including ordered input and output sorts) are enumerable from at most $2f(r+1)$ cut positions in an observed word. These facts verify Assumptions 1–3. □

Proof of Result 3.8. This is the rule-witness argument of [15, Lemma 3], applied to the concrete Link/Const/Comp presentation. For each typed rule put $\vec{z} = \rho(\iota(Y_1), \ldots, \iota(Y_q))$. The two witnesses supply Link $[\iota(X)] \to [\vec{z}]$, followed by Const if $q = 0$ or Comp of rank $q \leq r_G$ otherwise. The latter also belongs to the uncapped grammar, since $q \leq \|\vec{z}\|_1 \leq \|K\|_+$. Induction on a source derivation thus simulates $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ from $[\iota(X)]$; $\chi(X) = \Box_1$ at a successful start link supplies Start. Apply Proposition 3.5(iii). □

Proof of Result 3.10. If $L \cap \Sigma^+ = \emptyset$ then $K = L \in \{\emptyset, \{\varepsilon\}\}$ and the claim is immediate from Definition 3.3. Otherwise $\mathcal{F}_L \subseteq L$ is finite by the remark after Definition 3.6. Proposition 3.8 with $\iota = \omega$ (admissible by Proposition 3.5(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, and Proposition B.6 gives $L(\mathcal{B}(K)) \subseteq L$, for both operators. The hypothesis contains ε iff $\varepsilon \in K$, iff $\varepsilon \in L$, because $\mathcal{F}_L \subseteq K \subseteq L$. Rank-preserving normalization gives the last sentence. □

C Auxiliary Proofs for Typed Witness Construction

The statements used for quantitative witness bounds remain in Section 3.4; the routine verification details are recorded here.

Proof of Proposition 3.5. At every rule application, the monoid morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1), \ldots, \widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on derivation height gives (i) and the unique lifting (ii). Erasing indices gives $L(G^{\widehat{s}}) \subseteq L(G)$, (ii) the converse, and trimming keeps every rule of every successful derivation. □

Lemma C.1 (Monoid implementations)**.** *Let $h_0 : \Sigma^* \to M_0$ be a finite-monoid morphism, $S_0 := h_0(\Sigma^+)$ and $s_0 := h_0|_{\Sigma^+}$.*

(i) *For $1 \leq d \leq f$ and $\vec{x}, \vec{y} \in (\Sigma^+)^d$, the equalities $h_0^{(d)}(\vec{x}) = h_0^{(d)}(\vec{y})$, $s_0^{(d)}(\vec{x}) = s_0^{(d)}(\vec{y})$ and $\widehat{s_0}^{(d)}(\vec{x}) = \widehat{s_0}^{(d)}(\vec{y})$ are equivalent. Hence $\mathcal{C}_{f,h_0}^{\mathrm{mcf}} = \mathcal{C}_{f,s_0}^{\mathrm{mcf}}$, and the reconstruction operators built with h_0 and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under h_0 and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen for both.*

Proof of Lemma C.1. (i) The three maps agree on Σ^+ as maps into S_0, and the class and the Link guard compare types of nonempty components only. (ii) S_0 is a subsemigroup of M_0 and of $S_0^\sharp$ with the same multiplication. By λ-freeness and Proposition 3.5(i), every typed nonterminal occurring in a successful derivation under either morphism is some $A_{\mathbf{t}}$ with $\mathbf{t} \in S_0^{\mathrm{fan}(A)}$, and the start links used are $\widetilde{S} \to S_{(\eta)}$ with $\eta \in S_0$. For child indices in S_0, every component α_i of a good template is a nonempty word whose letters and variables are evaluated in S_0 identically under both morphisms, so both give the same out_ρ. Thus the two full refinements contain the same rules among S_0-indexed symbols, which carry all successful derivations. □

Lemma C.2 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \Box_1$.*

Proof of Lemma C.2. Take a successful typed derivation containing X, and follow the path from the root to an occurrence of X, starting with $\Box_1$. At each rule on the path apply Lemma 3.1 with the (nonempty) yields of the sibling subderivations fixed; this gives $\chi(X) \in \mathcal{E}_d$. The rules on the path depend only on the label X, so replacing the subtree of X by any X-derivation yields a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, use the path of length zero. □

Lemma C.3 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted $[X]$.*

Proof of Lemma C.3. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.5(i) gives equal types and $\chi(X)$ accepts both; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

D Details of the Good Normalization

Proof of Proposition 3.9. Trim; apply the nondeleting construction (f3) of [29, proof of Lemma 2.2] by retaining subsets of output components; decorate nonterminals by component permutations; decorate them by realizable nonempty-component patterns and project away empty components (this removes ε and makes all templates λ-free); finally apply Yoshinaka's nonmerging conversion [14, Section 2.3, Lemma 1]. No step increases fan-out or rule rank, and with $f, r_{\max}$ fixed each step creates $O_{f,r_{\max}}(1)$ copies of each nonterminal and rule (at most 2^f, $f!$ and $2^{fr_{\max}}$ decorations). The derivation correspondences and the size bookkeeping are given in Appendix D; cf. also [31, Section 2.1] and [14, Section 2.3, after Example 2]. □

We record the four conversions used in Proposition 3.9 once, together with the tuple-language identity that each satisfies. Both the fixed-parameter size bound of that proposition and the thickness transfer of Proposition D.2 are read off from these identities. Semantic correctness of the nondeleting and nonmerging conversions is standard [14, 29, 31]; Kanazawa's existence statement [31, Section 2.1] does not by itself give the size or thickness bounds, which are therefore derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ retains the components indexed by I in increasing order, $\sigma\vec{w}$ permutes components, and $\mu(\vec{w})$ concatenates consecutive blocks of components; thus $\|\pi_I(\vec{w})\|_1 \le \|\vec{w}\|_1$ and $\|\sigma\vec{w}\|_1 = \|\mu(\vec{w})\|_1 = \|\vec{w}\|_1$.

Lemma D.1 (Stagewise tuple languages)**.** *Fix f and $r_{\max}$, and let G_0 be a reduced f-MCFG of rule rank at most $r_{\max}$. Each of the following conversions preserves linearity, does not increase fan-out or rule rank, and multiplies the number of nonterminal and rule copies by a factor depending only on f and $r_{\max}$, without enlarging any template.*

(a) Nondeletion *(the construction (f3) in the proof of [29, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the set of components of B_j used by the I-indexed outputs of ρ; keep these outputs, rename their variables, use $B_j^{J_j}$ when $J_j \neq \emptyset$ and omit B_j otherwise. The resulting rules are nondeleting, and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$; the order in which a permuted parent template uses a child's variables determines the child's permutation. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let $\mathsf{Pat}(A)$ be the set of nonempty-component patterns of tuples in L_A, computed by finite saturation. For realizable child patterns, replace variables of absent components by ε, delete empty parent components and omit all-empty children; introduce A^I for each nonempty $I \in \mathsf{Pat}(A)$. Then L_{A^I} is the set of π_I-projections of the tuples of L_A with pattern I, the result is λ-free, and $S^{\{1\}}$ generates the nonempty part of the language. If every derived component is already nonempty, this stage is the identity.*

(d) Nonmerging *(the construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition μ of its fan-out; in a merged parent template, maximal adjacent runs of consecutive variables of a child determine that child's merge. The merge-decorated nonterminal generates the* image $L_{A^\mu} = \mu(L_A)$.

Stages (b)–(d) preserve the conditions established by earlier stages.

Proof. (a) For arbitrary child tuples, $\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$, so induction on derivations gives $\subseteq$. For $\supseteq$, lift every retained child tuple by the induction hypothesis and supply a terminal derivation of each omitted child; this uses that G_0 is reduced, hence every omitted child is productive. Without prior trimming only $\pi_I(L_A) \subseteq L_{A^I}$ would be automatic. There are at most 2^f copies of each nonterminal, and I determines the J_j, so at most 2^f copies of each rule; omitting children cannot increase rank.

(b) Linearity and nondeletion make the induced child permutations well defined; induction in both directions identifies the language with σL_A. There are at most $f!$ copies, no child is duplicated, and total length, λ-freeness, fan-out and rank are preserved.

(c) Retained variables still occur once and in order, so nondeletion, nonpermutation, fan-out and rank are preserved. Induction on derivations gives the projection identity; omitted all-empty children are restored by terminal derivations realizing their all-empty pattern. There are at most 2^f patterns per nonterminal and $2^{fr_{\max}}$ choices per rule, and projection only shortens templates.

(d) Nonpermutation makes the maximal runs compatible with each child's component order, and nondeletion makes them cover all its components. The merged template satisfies $\rho^\mu(\mu_1(\vec{v}_1), \ldots, \mu_q(\vec{v}_q)) = \mu(\rho(\vec{v}_1, \ldots, \vec{v}_q))$, and induction gives the image equality. No inverse-image claim is made, because the merge map need not be injective. Concatenation preserves total length, λ-freeness, nonpermutation, fan-out and rank; for fixed dimension and rank the conversion has linear size [14, Section 2.3, after Example 2]. □

Proof of the size bound in Proposition 3.9. Apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the encoding by a factor depending only on $f, r_{\max}$ and trimming can only decrease it, so $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$, and $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ by (c).

Thickness transfer for λ-free presentations

Proposition D.2 (Thickness under good-form conversion)**.** *Fix f and $r_{\max}$. Let G_{in} be a reduced λ-free f-MCFG of maximum rule rank at most $r_{\max}$, possibly with deleting rules, and suppose $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$. The good normalization of Proposition 3.9 yields an equivalent reduced good grammar G with the same fan-out and rule-rank bounds such that*

$$\|G\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|), \qquad t_G \le t_{G_{\mathrm{in}}}, \qquad \theta_G = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\}).$$

Proof. Because G_{in} is λ-free, every component of every derived tuple is nonempty, by induction on derivations; this persists through (a) and (b), so stage (c) is the identity. By Lemma D.1, every nonterminal X of the final grammar is associated with an original nonterminal A such that

$$L_X(G) = \mu\big(\sigma\pi_I(L_A(G_{\mathrm{in}}))\big), \qquad \min_{\vec{u} \in L_X(G)} \|\vec{u}\|_1 \le \min_{\vec{w} \in L_A(G_{\mathrm{in}})} \|\vec{w}\|_1.$$

Trimming leaves the tuple languages of surviving nonterminals unchanged, so $t_G \le t_{G_{\mathrm{in}}}$, and the size bound was shown above. Finally Lemma 4.1(iii) gives $\theta_G \le 2\|G\| \max\{1, t_G\} = O_{f,r_{\max}}(\|G_{\mathrm{in}}\| \max\{1, t_{G_{\mathrm{in}}}\})$. □

Thus deleting rules do not by themselves obstruct polynomial thickness transfer from a reduced λ-free presentation; genuine empty components are the problematic step.

Proposition D.3 (Empty-component elimination can blow up thickness)**.** *For every $n \ge 1$ there is a reduced fan-out-two, rule-rank-two MCFG E_n of size $O(n)$ with $t_{E_n} \le 2$ and $\theta_{E_n} = O(n)$ such that the empty-component elimination stage of Proposition 3.9 contains a retained nonterminal and a retained rule of thickness at least 2^n.*

Proof. Let E_n have fan-out-one S, fan-out-two $A_0, \dots, A_n$, and

$$A_0 \to (\mathtt{a}, \varepsilon), \qquad A_i \to (\varepsilon, \mathtt{c}) \mid (\xi_1^1\xi_2^1, \xi_1^2\xi_2^2)(A_{i-1}, A_{i-1}) \quad (1 \le i \le n), \qquad S \to (\xi_1^1 \mathtt{d}\, \xi_1^2)(A_n).$$

It is reduced, nondeleting, nonpermuting, and nonmerging. Short side derivations give $t_{E_n} \le 2$, and placing any chosen rule on one branch of the binary spine while completing siblings minimally gives $\theta_{E_n} = O(n)$. If an A_i-tuple has nonempty first and empty second component, the rule $(\varepsilon, \mathtt{c})$ cannot occur in it; inductively the shortest such tuple is $(\mathtt{a}^{2^i}, \varepsilon)$. Thus pattern $\{1\}$ for A_n survives, and the projected start rule $S \to \xi_1^1 \mathtt{d}\, (A_n^{\{1\}})$ succeeds only with a child of length at least 2^n. □

This is the obstruction of Proposition 4.5: finite state refinement can force exponentially long shortest witnesses inside one refined state.

E Further Forcing-Depth and Finite-Target Results

This appendix collects operator-independent distinguishing-pair criteria, abelian-coset certificates and finite-target marker-spectrum restrictions. They constrain prospective high-fan-out lower bounds but are not used to prove the main positive upper bounds of Section 4.

E.1 Distinguishing Pairs and Operator-Independent Data Lower Bounds

The following distinguishing-pair argument is elementary for set-driven learning and introduces no new learning algorithm. The companion CFG paper uses a related argument [25, Proposition 6.4(iv)]. For MCFGs, we make explicit that the relevant filters preserve fan-out and rule rank.

Lemma E.1 (Regular intersections preserve fan-out and rule rank)**.** *Let $f, r \geq 1$. If $L \in \mathsf{MCFL}(f, r)$ and $R \subseteq \Sigma^*$ is regular, then $L \cap R \in \mathsf{MCFL}(f, r)$.*

Proof. Let $\mathcal{D} = (Q, \Sigma, \delta, q_0, F)$ be a DFA for R. Refine each nonterminal A of fan-out $d \leq f$ in a linear nondeleting MCFG for L by d pairs of DFA states, one entry and exit state for each output component. For each production retain exactly the finitely many assignments of states for which all terminal constants and occurrences of child components are compatible with the left-to-right DFA transitions. Each child component occurs exactly once, so the constraints are local to the production; neither the number of output components nor the number of children increases. At the start allow only pairs (q_0, q) with $q \in F$. The resulting finite grammar generates $L \cap R$ without increasing fan-out or rule rank. This is the standard regular-intersection product construction for MCFGs [29], with its rank bound made explicit. □

Lemma E.2 (Forcing by a distinguishing pair)**.** *Let $\mathcal{C}$ be a language class and let* Op *be a set-driven operator having a characteristic sample for every language in $\mathcal{C}$. If $L', L \in \mathcal{C}$ and $L' \subsetneq L$, then every characteristic sample K of L for* Op *meets $L \setminus L'$.*

Proof. Otherwise $K \subseteq L'$. Choose a characteristic sample K' for L'. The finite sample $K \cup K' \subseteq L'$ contains both characteristic samples, so $\mathrm{Op}(K \cup K')$ would generate both L and L', a contradiction. □

Corollary E.3 (Forcing by fibres and finite-index cosets)**.** *Fix $f, r \geq 1$ and a finite-semigroup typing s, and put $\mathcal{C} := \mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r)$. Let $L \in \mathcal{C}$ and let R be one of the following:*

(i) $R = \widehat{s}^{-1}(F)$ *with* $F \subseteq \mathsf{S}^\sharp$*;*

(ii) $R = \varphi^{-1}(\gamma + \mathsf{H})$*, where* $\varphi : \Sigma^* \to \mathsf{A}$ *is a morphism into an abelian group,* $\mathsf{H} \leq \mathsf{A}$ *has* finite index*, and* $\gamma \in \mathsf{A}$*.*

If $L \setminus R \neq \emptyset$, every characteristic sample K of L for every set-driven operator having characteristic samples throughout $\mathcal{C}$ satisfies $K \cap (L \setminus R) \neq \emptyset$. In particular, K meets every nonempty $\widehat{s}$-type fibre of L.

Proof. In (i), R is regular as an inverse image under a finite monoid morphism. In (ii), R is regular since it is recognized by the finite quotient A/H. For the finite type-fibre filter, equality of componentwise s-values ensures that every admissible sentence context gives the same filter value, so intersection preserves (f, s)-tuple substitutability. For the finite-index abelian-coset filter, the same preservation follows from Proposition 2.9(ii). By Lemma E.1, it is also in $\mathsf{MCFL}(f, r)$. Therefore $L' = L \cap R \in \mathcal{C}$ is a proper sublanguage, and Lemma E.2 applies. To force a word in a nonempty type fibre of type μ, use $R = \widehat{s}^{-1}(\mathsf{S}^\sharp \setminus \{\mu\})$. □

Let $\mathcal{F}_s$ be the union of the two families of filters above and define their *forcing depth* for $L \in \mathcal{C}$ by

$$\Delta_s(L) := \sup_{\substack{R \in \mathcal{F}_s \\ L \setminus R \neq \emptyset}} \min_{w \in L \setminus R} (|w| + 1),$$

where the supremum of the empty family is defined to be zero. The corollary gives $\|K\|_+ \geq \Delta_s(L)$ for every characteristic sample of the operators in question. Theorem 3.10 provides such an operator under a fixed rank cap, so $\Delta_s(L)$ is finite for every fixed target $L \in \mathcal{C}$. Superpolynomial forcing depth along a target family would provide an operator-independent lower bound; no converse is claimed.

Open Problem E.4 (Operator-independent higher-fan-out lower bounds). Fix $f \geq 2$ and $r \geq 2$. Does there exist a fixed finite-semigroup typing s and a family of reduced good fan-out-$\leq f$, rank-$\leq r$ grammars G_n generating targets in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, with $\|G_n\|\bar{\theta}_{G_n} = n^{O(1)}$, such that for *every* set-driven operator having characteristic samples on all of $\mathcal{C}$, every characteristic sample of $L(G_n)$ has superpolynomial total encoded length? The condition $\Delta_s(L(G_n)) = n^{\omega(1)}$ would suffice; it is not known to be necessary.

E.2 Short coset certificates and obstructions for finite targets

We record two limitations on lower-bound constructions. The first restricts *which competitors* can force large positive samples; the second restricts *which finite grammars* can hide very long examples behind short rule witnesses. Neither settles Open Problems 6.4 and E.4.

For a tuple $\vec{u}$ let $\Psi(\vec{u})$ denote the sum of the Parikh vectors of its components, and write $\psi(w)$ for the Parikh vector of a word. The following identity makes explicit the rule-difference mechanism used in Theorem 4.39.

Proposition E.5 (Short Parikh-difference certificates)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of rule rank at most r, with $\emptyset \neq L(G) \subseteq \Sigma^+$, and put $\theta = \theta_G$. For each $A \in V$ choose a minimum-length tuple $\omega_A \in L_A(G)$. For a rule $R : A \to \rho(B_1, \ldots, B_q)$ let c_R count the explicit terminals in ρ and put $\delta_R = c_R + \sum_{j=1}^{q} \Psi(\omega_{B_j}) - \Psi(\omega_A) \in \mathbb{Z}^{\Sigma}$. Then*

$$\langle \psi(u) - \psi(v) : u, v \in L(G) \rangle_{\mathbb{Z}} = \langle \delta_R : R \in P \rangle_{\mathbb{Z}}.$$

Each δ_R is the difference of the Parikh vectors of two positive words of length at most $(r+2)\theta$. Moreover, $\|\delta_R\|_1 \leq (r+2)\theta$.

Proof. For a derivation tree T rooted at A and yielding $\vec{u}$, expand $\sum_{v \in T} \delta_{R(v)}$. Each non-root reference vector $\Psi(\omega_B)$ cancels between its own node and its parent's rule; linearity and nondeletion give $\Psi(\vec{u}) - \Psi(\omega_A) = \sum_{v \in T} \delta_{R(v)}$. For the start symbol this proves one inclusion.

Conversely, Lemma 4.1(iv) supplies for each A an admissible context E_A with at most θ fixed terminals that accepts every tuple in $L_A(G)$. Both $u_R = E_A[\omega_A]$ and $v_R = E_A[\rho(\omega_{B_1}, \ldots, \omega_{B_q})]$ are in $L(G)$, and $\psi(v_R) - \psi(u_R) = \delta_R$. Lemma 4.1(i)–(ii) bounds each reference tuple and each rule's explicit terminal material by θ. Thus $|u_R| \leq 2\theta$, $|v_R| \leq (q+2)\theta$, and the triangle inequality applied directly to the definition of δ_R gives $\|\delta_R\|_1 \leq (q+2)\theta$. □

Corollary E.6 (Uniform certificates for abelian-coset competitors)**.** *Under the assumptions of Proposition E.5, there is a single set $T_G \subseteq L(G)$ of at most $|V| + |P|$ words with $\|T_G\|_+ \leq (|V| + |P|)(1 + (r+2)\theta_G)$, such that for* every *morphism $\varphi : \Sigma^* \to \mathsf{A}$ into an abelian group, subgroup $H \leq \mathsf{A}$ and coset $\gamma + H$,*

$$T_G \subseteq \varphi^{-1}(\gamma + H) \quad \Longrightarrow \quad L(G) \subseteq \varphi^{-1}(\gamma + H).$$

For a fixed finite typing s put $M_s(L) = \max_{t \in \widehat{s}(L)} \min\{|w| + 1 : w \in L, \widehat{s}(w) = t\}$. For the filter family $\mathcal{F}_s$ defined above, when $L = L(G) \subseteq \Sigma^+$ belongs to the corresponding typed class,

$$M_s(L) \leq \Delta_s(L) \leq \max\{M_s(L), (r+2)\theta_G + 1\}.$$

All proper competitors $L \cap R$ with $R \in \mathcal{F}_s$ have their difference sets simultaneously met by some positive set of weight at most $(|V| + |P|)(1 + (r+2)\theta_G) + |\mathsf{S}|M_s(L)$.

Proof. Take T_G to contain $E_A[\omega_A]$ for each A, and $E_A[\rho(\omega_{B_1}, \ldots, \omega_{B_q})]$ for each rule R. If all these words lie in a coset, every δ_R maps into H; the preceding lattice identity then puts every word of $L(G)$ into that coset. Hence each coset filter which excludes a target word is violated by a member of T_G, of length at most $(r+2)\theta_G$. For a type filter which excludes a realized type, a shortest word of that type has encoded length at most $M_s(L)$; conversely, excluding a type achieving $M_s(L)$ gives the lower bound. Adjoining one shortest word for each realized type to T_G proves the simultaneous bound. □

The uniform set above need not be a characteristic sample of any learner. Thus the corollary excludes superpolynomial *hitting-set* arguments based only on these abelian-coset competitors when the original grammar thickness is polynomial; it does not exclude other competitors or general operator-independent lower bounds.

We turn to finite languages. A *componentwise factor embedding* $\vec{x} \preceq_{\mathrm{fac}} \vec{y}$ means that each x_i is a contiguous factor of y_i.

Lemma E.7 (Finite-target tuple-nesting obstruction)**.** *Let L be finite and (f, s)-tuple-substitutable. If nonempty d-tuples $\vec{x}, \vec{y}$, $d \le f$, have equal componentwise types and a common accepting admissible context, then $\vec{x} \preceq_{\mathrm{fac}} \vec{y}$ implies $\|\vec{x}\|_1 = \|\vec{y}\|_1$. In particular, if G is a reduced good grammar for L, each componentwise-type fibre of $L_A(G)$ is an antichain under proper componentwise factor embedding.*

Proof. Suppose $y_i = p_i x_i q_i$ and $h = \sum_i (|p_i| + |q_i|) > 0$. Equality of the full tuple distributions follows from substitutability. Write $T(\vec{z}) = (p_i z_i q_i)_i$ and let E be a common accepting context. Every $E[T^k(\,\cdot\,)]$ is still an admissible d-hole context: the original nonempty separators remain in place. Inductively, the equality of distributions forces $E[T^k(\vec{x})] \in L$ for all $k \ge 0$, producing words of strictly increasing lengths, a contradiction. For grammar yields use the common successful context E_A from Lemma 4.1(iv). □

For the next result fix $\Sigma = \{\mathtt{a}, \#\}$ and the non-local-group two-element semilattice typing $s(\mathtt{a}) = 0$, $s(\#) = 1$, with multiplication max. The positive type-zero fibre is $L \cap \mathtt{a}^+$.

Lemma E.8 (Transport of short unary mass)**.** *Suppose L is $(2, s)$-tuple-substitutable and $N = \min\{k \ge 1 : \mathtt{a}^k \in L\}$ exists. For $2 \le m < N$, the tuples $(\mathtt{a}^i, \mathtt{a}^{m-i})$, $1 \le i < m$, have the same complete two-hole distribution. Consequently, if $\mathtt{a}^p z \mathtt{a}^{m-p} \in L$ for any nonempty z and $1 \le p < m$, then $\mathtt{a}^i z \mathtt{a}^{m-i} \in L$ for all $1 \le i < m$.*

Proof. All the displayed tuples have type $(0, 0)$, and the admissible context $\Box_1 \mathtt{a} \Box_2 \mathtt{a}^{N-m-1}$ accepts every one, producing $\mathtt{a}^N$. Apply two-dimensional substitutability, then use $\Box_1 z \Box_2$. □

Theorem E.9 (Linear short-fibre bound with an interior marker)**.** *Let G be a reduced good MCFG of rule rank at most r with finite nonempty $L = L(G) \subseteq \mathtt{a}^+ \cup \mathtt{a}^+ \# \mathtt{a}^+$. If L is $(2, s)$-tuple-substitutable, then*

$$M_s(L) \le (r+2)\theta_G + 2.$$

This holds for arbitrary fan-out, provided the relevant two-hole substitutability condition is imposed.

Proof. For a finite $(1, s)$-substitutable language, two accepted words of the same type cannot properly contain one another as contiguous factors: apply Lemma E.7 to the identity context. In particular, $L \cap \mathtt{a}^+$ has at most one word. If only one type is realized, its shortest word has length at most θ_G; suppose henceforth that both are realized, with type-zero word $\mathtt{a}^N$. Some rule writes

#, and its shortest successful use gives a type-one word of length at most θ_G. If $N \le \theta_G$ the conclusion follows. Otherwise write this short word as $\mathtt{a}^p\#\mathtt{a}^q$, $p, q \ge 1$, and put $m = p + q < N$. If $m \ge 3$, Lemma E.8 produces every interior split of total unary mass m. Comparing $\#\mathtt{a}^{m-1}$ and $\mathtt{a}\#\mathtt{a}^{m-2}$ under the common context $\mathtt{a}\Box_1$, and then using $\Box_1\mathtt{a}$, forces $\#\mathtt{a}^m \in L$, contrary to the interior-marker hypothesis. Thus $m = 2$ and the short word is $\mathtt{a}\#\mathtt{a}$. Every other word of type one properly contains this word as a factor, which contradicts Lemma E.7. Hence $L = \{\mathtt{a}^N, \mathtt{a}\#\mathtt{a}\}$.

The Parikh-difference group is generated by $(N-2, -1)$ and is nontrivial. Proposition E.5 therefore supplies a nonzero rule defect. As its defining positive words belong to the two-word language, the defect is $\pm(N-2, -1)$, of norm $N-1$. Consequently $N - 1 \le (r+2)\theta_G$, giving the claimed bound including encoded word cost. □

The finite-language hypothesis above is essential to the factor argument; the interior-marker hypothesis excludes boundary markers. Neither assumption follows from general two-dimensional substitutability.

Theorem E.10 (Length spectrum of a finite good MCFG)**.** *Let G be a reduced good MCFG of rule rank at most r with nonempty finite language $L(G)$. Put $\theta = \theta_G$ and $t = |\{|w| : w \in L(G)\}|$. Then*

$$\max_{w \in L(G)} |w| \le (r+1)|V|\,\theta\,\max\{1, t-1\}.$$

This bound does not assume any semigroup typing or substitutability.

Proof. Choose, for each accepted word, a successful derivation with the fewest nodes. No root-to-leaf path of such a derivation repeats a nonterminal. Indeed, a return path from A to A either introduces some terminal or a productive side-child, in which case repeating it makes $L(G)$ infinite by linearity and nondeletion; or it introduces nothing, in which case every step is a unary, terminal-free variable redistribution. Since each component remains nonempty, and the endpoints have the same fan-out, the composite redistributes the components bijectively. The nonpermuting condition makes that bijection the identity, so the zero-growth return can be removed, contradicting minimality. Thus every such derivation has depth at most $|V|$.

Call a node heavy if the total length of its tuple yield exceeds θ. Every nonterminal has a yield of total length at most θ by Lemma 4.1(ii). If there are k pairwise incomparable heavy nodes, replace these subtrees, one at a time, by short yields. Each replacement strictly decreases the final length, so the resulting $k+1$ accepted lengths are distinct. Hence $k \le t-1$. The heavy nodes form an ancestor-closed subtree of depth at most $|V|$, and therefore number at most $|V|(t-1)$. If the root is light, $|w| \le \theta$. Otherwise every terminal is written either by a heavy node itself (at most θ terminals per node) or by one of its at most r light children (at most θ terminals in each). It follows that $|w| \le (r+1)\theta|V|(t-1)$, as required. □

Thus, if a family of finite good grammars has polynomial original size and thickness but accepts a superpolynomial-length word, it must have superpolynomially many distinct accepted lengths. This necessary condition alone does not settle either higher-fan-out open problem.

E.3 Rare unary fibres under semilattice typing

Fix $\Sigma = \{\mathtt{a}, \#\}$ and $s : \Sigma^+ \to (\{0,1\}, \max)$ with $s(\mathtt{a}) = 0$ and $s(\#) = 1$. The following two results distinguish a semantic obstruction from a constraint imposed by the original good grammar.

Definition E.11 (Unary mass-transport saturation)**.** A language $D \subseteq \Sigma^+$ containing a marker in every word has *unary mass-transport saturation* (UMTS) if, for all $m \ge 2$ and $1 \le i, j < m$,

$$\mathcal{D}^{(2)}_D(\mathtt{a}^i, \mathtt{a}^{m-i}) = \mathcal{D}^{(2)}_D(\mathtt{a}^j, \mathtt{a}^{m-j}).$$

Theorem E.12 (Finite-marker unary adjunction)**.** *Let $D \subseteq \Sigma^+$ be finite and nonempty, with every word containing #, and put $\ell = \max_{w \in D} |w|$. For $N \geq \ell + 2$, let $L_N = D \cup \{\mathtt{a}^N\}$. Then*

$$L_N \in \mathcal{C}_{2,s}^{\mathrm{mcf}} \quad \Longleftrightarrow \quad D \in \mathcal{C}_{2,s}^{\mathrm{mcf}} \text{ and } D \text{ satisfies UMTS.}$$

Proof. If L_N is substitutable, so is D: for equal-type tuples, membership of one completion in D forces the other completion, if accepted by L_N, to contain a marker as well. If two equal-total-mass unary pairs had different D-distributions, their accepting D-context would imply $m \leq \ell$; the admissible context $\square_1\mathtt{a}\square_2\mathtt{a}^{N-m-1}$ accepts both pairs in $\{\mathtt{a}^N\}$, contradicting substitutability. Hence UMTS holds.

Conversely let D be substitutable and satisfy UMTS. Take equal-component-type tuples $\mathbf{x}, \mathbf{y}$ of arity $d \leq 2$ with a common accepting context in L_N. Both completions have the same marker-presence type. If both are $\mathtt{a}^N$, the tuples are unary and have equal total length; when $d = 1$ they coincide, and when $d = 2$ UMTS makes their D-distributions equal. Membership in $\{\mathtt{a}^N\}$ depends only on total unary length. If both common completions belong to D and a tuple component contains a marker, neither tuple can be accepted in the unary singleton, so substitutability of D suffices. For two all-unary tuples accepted together by a D-context, equal total lengths give the same conclusion by UMTS (or, for $d = 1$, identity of the tuples). If their total lengths are $m < m'$, substitutability of D for $d = 1$ or UMTS for $d = 2$ makes that context accept the canonical tuples $(\mathtt{a}, \ldots, \mathtt{a}, \mathtt{a}^{m-d+1})$ and $(\mathtt{a}, \ldots, \mathtt{a}, \mathtt{a}^{m'-d+1})$. The first is a proper componentwise factor of the second, contradicting the finite-target tuple-nesting obstruction (Lemma E.7). Thus unequal masses cannot arise, and the complete distributions agree. □

Example E.13 (An order-sensitive rare-unary semantic witness)*.* For every $N \geq 7$, the language

$$L_N = \{\mathtt{a}^N, \mathtt{aa}\#\#, \#\#\mathtt{a}\#, \mathtt{a}\#\mathtt{a}\#\mathtt{a}\}$$

is $(2, s)$-tuple-substitutable but not Parikh-closed. Indeed, an enumeration of the admissible one- and two-hole success contexts of the three marker words shows that their language D is $(2, s)$-tuple-substitutable and UMTS: the only accepted all-unary two-tuple is $(\mathtt{a}, \mathtt{a})$. Theorem E.12 applies. The context $\mathtt{a}\square_1\#$ distinguishes $\mathtt{a}\#$ from $\#\mathtt{a}$. Moreover, the differences of the marker-word Parikh vectors $(2, 2), (1, 3), (3, 2)$ generate $\mathbb{Z}^2$. The three distinct accepted lengths are $4, 5, N$; hence every reduced good rank-$\leq r$ grammar H for L_N obeys $N \leq 2(r + 1)|V_H|\theta_H$ by Theorem E.10. Thus this semantic witness cannot have polynomial original size and thickness with N exponential.

Theorem E.14 (Short-step marker-count spectrum)**.** *Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a reduced good MCFG of rule rank at most r, with $\theta = \theta_G \geq 1$ and $v = |V|$. Suppose $L(G) \cap \mathtt{a}^+ \neq \varnothing$, and put*

$$N = \min\{m \geq 1 : \mathtt{a}^m \in L(G)\}, \qquad H_{\#}(L) = \{|w|_{\#} : w \in L(G)\}.$$

If $N > \theta$, then some integers b, k satisfy

$$1 \leq b \leq \theta, \qquad k \geq \left\lceil \frac{N}{(r+1)v^2\theta} \right\rceil, \qquad \{0, b, 2b, \ldots, kb\} \subseteq H_{\#}(L).$$

In addition, if $h = |H_{\#}(L)|$ is finite, then

$$N \leq (r + 1)v\theta \max\{1, h - 1\}.$$

No finiteness, fan-out bound, or substitutability assumption on the target language is needed.

Proof. Choose a derivation tree T of the shortest unary word $\mathtt{a}^N$ with the fewest nodes. All subtree yields are unary, by linearity and nondeletion. No root-to-leaf path repeats a nonterminal: replacing an upper occurrence of A by its lower A-subtree gives an accepted unary word of length at most N using fewer nodes, a contradiction. Thus T has depth at most v.

Call a node *heavy* if its tuple yield has total length greater than θ, and write h_T for the number of heavy nodes. They form an ancestor-closed set. Each heavy node writes at most θ terminals and has at most r light children, each of yield length at most θ. Consequently

$$N \leq (r+1)\theta h_T.$$

Some depth of T contains at least h_T/v heavy nodes, of which at least h_T/v^2 have the same nonterminal A. These occurrences are pairwise incomparable. By Lemma 4.1(ii), A derives a tuple ω_A of total length at most θ. It cannot be unary, since replacing one of the heavy A-subtrees by it would yield a unary accepted word shorter than $\mathtt{a}^N$. Hence ω_A has b markers, $1 \leq b \leq \theta$. Replacing any j of the selected occurrences by ω_A, independently, produces a successful word with precisely jb markers. The bound on k follows.

For the finite-spectrum claim, at any fixed depth let there be q heavy nodes. Replace each by a short tuple for its nonterminal, in succession. Every replacement strictly increases the marker count, so $q+1$ distinct marker counts are accepted and $q \leq h-1$. There are at most v depths; thus $h_T \leq v(h-1)$. The bound follows, including the case $N \leq \theta$. □

Corollary E.15 (Sparse marker-count obstruction)**.** *Let $B_\theta(L)$ be the supremum of integers $k \geq 1$ for which $\{0, b, \ldots, kb\} \subseteq H_\#(L)$ for some $1 \leq b \leq \theta$, and put $B_\theta(L) = 0$ if there is no such k. If $B_\theta(L) < \infty$, then*

$$N \leq (r+1)v^2\theta \max\{1, B_\theta(L)\}.$$

Therefore polynomial original grammar size and thickness together with superpolynomial N require a superpolynomially rich marker-count spectrum, including arbitrarily long progressions of step at most θ.

Proof. For $N > \theta$ apply the progression in the preceding theorem; for $N \leq \theta$ the estimate is immediate. □

These constraints do not settle the higher-fan-out characteristic-data lower-bound problems: the example isolates a semantic possibility, whereas the marker-count theorem constrains its grammar presentations.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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